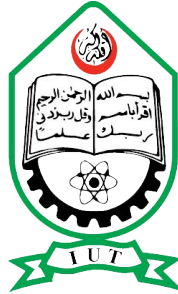


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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Uncertainty-Based Study of EV Charging Impacts Using PV, Wind and BESS in a Generic 9-Bus Microgrid with Harmonic Mitigation

A thesis submitted in partial fulfilment of the requirements
for the degree of Bachelor of Science in Electrical and Electronic Engineering

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DECLARATION

We hereby declare that this thesis, titled “*Uncertainty-Based Study of EV Charging Impacts Using PV, Wind and BESS in a Generic 9-Bus Microgrid with Harmonic Mitigation*”, is the outcome of our own work carried out under the supervision of Dr. Ashik Ahmed, Professor, Department of Electrical and Electronic Engineering, Islamic University of Technology.

We further declare that the network model, simulation studies, optimisation scripts and numerical results presented in this thesis were produced by us using DIgSILENT PowerFactory, and that all material drawn from other sources has been duly acknowledged through citation. No part of this work has been submitted elsewhere for the award of any degree or diploma.

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CERTIFICATE OF APPROVAL

This is to certify that the thesis titled “*Uncertainty-Based Study of EV Charging Impacts Using PV, Wind and BESS in a Generic 9-Bus Microgrid with Harmonic Mitigation*” submitted by Abrar Mubasshir Rahman (210021230), Mohammad Atahar Nur Faiz (210021257) and Zahir Sadik (210021228) has been carried out under my supervision. The work is original and has been prepared in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and Electronic Engineering.

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Finally, we thank our families and our classmates for their support over the two semesters of this project.

ABSTRACT

Electric vehicle charging is being added to distribution networks faster than those networks were designed to accommodate, and the chargers themselves are power-electronic loads that inject harmonic current rather than drawing a clean sinusoid. This thesis examines both sides of that problem in a generic nine-bus microgrid modelled in DIgSILENT PowerFactory: the load and generation uncertainty introduced by electric vehicle charging alongside photovoltaic and wind generation, and the harmonic distortion produced by the charging station itself.

An annual 8760-hour dataset was assembled from measured photovoltaic and wind generation for Bhola, Bangladesh, combined with an electric vehicle charging profile derived from NREL EVI-Pro Lite data. Input uncertainty was represented through a 300-run Monte Carlo study in which renewable output was perturbed by a first-order autoregressive noise process, preserving the hour-to-hour correlation that independent sampling would destroy. The resulting scenarios give a renewable penetration of 15.97 % to 17.38 % across runs.

Harmonic load flow was then performed with the electric vehicle charging station acting as the sole harmonic source. With the battery energy storage system connected but its harmonic compensation disabled, voltage total harmonic distortion reached 8.0781 % at Bus 8, with five of the nine buses at or above the 5 % benchmark adopted in this study and a system average of 5.3926 %.

The storage converter was then operated as an active harmonic compensator, injecting controlled fifth- and seventh-order current components while retaining its fundamental-frequency role. Particle swarm optimisation and differential evolution were implemented as PowerFactory Python scripts to tune the magnitude and phase of those components. Differential evolution reduced the worst-case distortion to 4.6866 % and the system average to 3.5907 %, bringing all nine buses below the benchmark; particle swarm optimisation reached 5.0285 % at Bus 5 and left that bus marginally non-compliant.

The most consequential result concerns converter sizing rather than the algorithms. Repeating the differential evolution study with a 1.0 MVA converter left two buses above the benchmark despite both harmonic magnitudes saturating at their upper bound, whereas the 1.5 MVA converter succeeded. Compensation capability was therefore limited by available converter current headroom, not by search performance. This finding sits in direct tension with published active-filter sizing practice, which suggests that harmonic duty requires only a small percentage margin, and the thesis examines why the

two disagree.

Keywords: electric vehicle charging, harmonic distortion, battery energy storage, active harmonic compensation, differential evolution, particle swarm optimisation, Monte Carlo simulation, DIgSILENT PowerFactory, microgrid, power quality.

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LIST OF ABBREVIATIONS

AR(1)	First-Order Autoregressive Process
BESS	Battery Energy Storage System
CR	Crossover Rate (differential evolution)
DE	Differential Evolution
DER	Distributed Energy Resource
EV	Electric Vehicle
HLF	Harmonic Load Flow
IEEE	Institute of Electrical and Electronics Engineers
IUT	Islamic University of Technology
LHS	Latin Hypercube Sampling
MCS	Monte Carlo Simulation
NREL	National Renewable Energy Laboratory
PCC	Point of Common Coupling
PSO	Particle Swarm Optimisation
PV	Photovoltaic
RE	Renewable Energy
SOC	State of Charge
THD	Total Harmonic Distortion
VFD	Variable Frequency Drive
WTG	Wind Turbine Generator

CHAPTER 1

INTRODUCTION

1.1 Background

Electricity networks were planned around a reasonably predictable relationship between generation and demand. Generation followed load, load followed the time of day, and the flow of power ran in one direction. Two developments have unsettled that arrangement at the same time. Distributed renewable generation has made supply intermittent at the point of connection, and electric vehicle (EV) charging has introduced a load that is large, mobile and concentrated in time. Neither change alone would be unmanageable. Together they make the question of what the network will actually be doing at a given hour much harder to answer.

Bangladesh is entering this transition now rather than in prospect. EV registration began formally in 2022, and public charging infrastructure has started to appear on major corridors [1], [2]. At the same time, large-scale renewable integration is being pursued for energy security, with the variability that implies [3]. A microgrid that serves EV charging from photovoltaic (PV) and wind generation is therefore not a hypothetical configuration for this country; it is a plausible near-term one.

1.2 EV Charging and Grid Integration

An EV charging station differs from conventional load in two respects that matter for this thesis. The first is magnitude and timing: charging demand is concentrated, poorly correlated with renewable availability, and varies substantially between days. The second is the nature of the load itself. A charger is a power-electronic converter. It draws current in a non-sinusoidal pattern determined by its switching behaviour, and that current propagates through the network as harmonic components at multiples of the fundamental frequency.

The consequences of this are well documented. Harmonic currents increase transformer losses and accelerate thermal ageing, distort the voltage waveform seen by neighbouring customers, and can cause maloperation of protection and metering equipment [4]. Because the distortion is carried by the network rather than confined to the charger, the

effect is felt at buses some distance from the charging station.

1.3 Renewable Energy Integration

PV and wind generation reduce dependence on conventional plant and can supply a meaningful share of charging energy. Hybrid PV–wind architectures are attractive for EV charging specifically because the two resources are imperfectly correlated, so their combination is less variable than either alone [5]. In the system studied here, renewable penetration across the generated scenarios lies between 15.97 % and 17.38 % of annual energy.

An important point for what follows is that PV and wind, as modelled in this study, are not harmonic sources. Their contribution is to the fundamental power balance and to load-flow conditions. This is a deliberate modelling decision rather than an oversight, and Section 3.7 sets out why.

1.4 Uncertainty in PV and Wind Generation

A recurring weakness in EV-integration studies is the treatment of renewable output as a fixed profile. A single deterministic year gives a single answer, and that answer says nothing about how the system behaves when conditions differ. Probabilistic approaches, in which repeated simulations are run over sampled inputs, give a distribution of outcomes instead [6], [7].

This thesis uses a 300-run Monte Carlo study in which PV and wind output are perturbed by a first-order autoregressive noise process. The choice of an autoregressive rather than independent perturbation matters: solar and wind output at consecutive hours are correlated, and independent hourly sampling produces a physically implausible series that understates sustained periods of low or high generation.

1.5 Battery Energy Storage Systems

A battery energy storage system (BESS) is normally justified on energy grounds: smoothing renewable variability, shaving demand peaks, and shifting energy in time. The converter that connects the battery to the network, however, is capable of more than fundamental-frequency power exchange. It is a controllable current source, and with appropriate control it can inject current at harmonic frequencies as well.

This is the mechanism the thesis exploits. The same converter that manages energy can be operated as an active harmonic compensator, injecting fifth- and seventh-order current components chosen to oppose those produced by the charging station. Prior work has established the principle, both for superposed BESS control strategies [8] and for bidirectional charging systems with harmonic compensation [9], [10]. What has received less attention is the capacity question: how much converter current the harmonic duty actually requires, and what happens when it is not available.

1.6 Harmonic Distortion from EV Charging

Total harmonic distortion (THD) of the bus voltage is the metric used throughout this thesis. For a bus voltage containing harmonic components V_h superimposed on a fundamental V_1 ,

$$\text{THD}_V = \frac{\sqrt{\sum_{h=2}^H V_h^2}}{V_1} \times 100 \% \quad (1.1)$$

where h is the harmonic order and H the highest order considered. A 5% voltage THD limit is widely used as a design benchmark and is adopted here as the compliance criterion. Section 2.8 discusses the basis for that figure and the care required before attributing it directly to a specific edition of IEEE Standard 519.

1.7 Problem Statement

An EV charging station connected to a renewable-supported microgrid produces voltage distortion that propagates across the network and, at several buses, exceeds the benchmark used for acceptable power quality. Adding PV, wind and a conventionally operated BESS does not by itself resolve this, because none of those elements injects the compensating harmonic current that would be required. A dedicated mitigation measure is therefore needed. The question this thesis addresses is whether the BESS converter already present for energy reasons can supply that mitigation, and what converter capability it requires to do so.

1.8 Motivation

Two considerations motivated this work. The first is practical: if an existing converter can perform harmonic compensation, the alternative of installing a separate active power filter is avoided, which matters in a cost-constrained deployment context. The second is methodological. Studies that report successful harmonic mitigation frequently do so without stating what limited the result, so it is not clear whether the reported performance was the best achievable or simply the first acceptable answer found. This thesis reports the binding constraint explicitly.

1.9 Research Gap

Reviewing the literature summarised in Chapter 2, three gaps are apparent.

First, EV-integration studies that model renewable uncertainty rigorously generally stop at load flow and do not extend to harmonic analysis, while harmonic studies of EV charging typically use deterministic operating points. The two strands are seldom joined.

Second, where BESS-based harmonic compensation is proposed, the converter is usually assumed to have sufficient current capability. The relationship between converter rating and achievable distortion reduction is rarely quantified for a specific network.

Third, published active-filter sizing guidance indicates that harmonic compensation requires only a small percentage margin over the rating set by other duties [11]. Whether this holds for a compensator placed at a single bus in a meshed network, rather than at the point of common coupling of the distorting load, has not been established.

1.10 Research Objectives

1. To build a nine-bus microgrid in DIgSILENT PowerFactory incorporating PV, wind, an EV charging station and a BESS, driven by an annual 8760-hour dataset derived from measured generation and real charging behaviour.
2. To represent input uncertainty through Monte Carlo sampling that preserves the temporal correlation of renewable output.
3. To quantify, by harmonic load flow, the voltage distortion produced at every bus by the EV charging station with the BESS present but its harmonic compensation disabled.
4. To model the BESS converter as an active harmonic compensator and to determine, by

optimisation, the harmonic current magnitudes and phases that minimise system-wide distortion.

5. To compare particle swarm optimisation and differential evolution for this task, and to identify what limits the achievable result.
6. To establish the relationship between BESS converter rating and achievable harmonic mitigation.

1.11 Research Questions

- RQ1. What level of voltage distortion does the EV charging station produce across the nine buses, and which buses are critical?
- RQ2. Can the BESS converter reduce distortion below the 5 % benchmark at every bus through fifth- and seventh-order current injection alone?
- RQ3. Which of the two optimisation algorithms performs better on this problem, and is the difference attributable to the algorithms themselves?
- RQ4. Is the achievable mitigation limited by the search procedure or by the physical capability of the converter?
- RQ5. How does converter rating affect the outcome?

1.12 Scope and Limitations

The study covers a single generic nine-bus microgrid modelled in PowerFactory, with harmonic analysis performed in the frequency domain by harmonic load flow.

Three boundaries should be stated at the outset. The harmonic study is carried out at one operating point rather than at every hour of the year; the annual dataset establishes the operating conditions and the uncertainty analysis, but the reported THD figures are not annual statistics. Compensation is applied at a single BESS location and considers the fifth and seventh harmonic orders only. Finally, the model is a frequency-domain representation; switching behaviour, converter control dynamics and thermal limits are outside its scope. Section 4.20 returns to these in detail.

1.13 Contributions

1. A PowerFactory nine-bus microgrid model with PV, wind, EV charging and BESS, driven by an annual dataset built from measured generation and real charging data for a Bangladeshi context.
2. A correctly posed harmonic baseline, in which the comparison isolates the effect of compensation rather than confounding it with the effect of adding distributed generation.
3. Two optimisation implementations that drive PowerFactory harmonic load flow directly through its Python interface, with the search operating on harmonic magnitude and phase simultaneously.
4. A converter-sizing result showing that mitigation capability was limited by available converter current rather than by the optimisation, obtained by repeating the study at a reduced rating.
5. An explicit account of where the sizing evidence in this work disagrees with published active-filter sizing practice.

1.14 Organisation of the Report

Chapter 2 reviews the relevant literature by theme and then sets out the methodology adopted. Chapter 3 documents the engineering work: the network model, the component representations, the dataset and uncertainty treatment, the harmonic study configuration, and the two optimisation implementations. Chapter 4 presents and interprets the results. Chapter 5 states the conclusions, the limitations that remain, and the work that would follow from this study.

CHAPTER 2

LITERATURE REVIEW AND METHODOLOGY

2.1 Introduction

This chapter has two parts. The first reviews existing work by theme, comparing methods and identifying where their limitations bear on the present study rather than summarising each paper in turn. The second sets out the methodology adopted here and explains the reasoning behind each methodological choice.

2.2 EV Charging Integration in Distribution Networks

Work on EV integration divides broadly into scheduling studies and impact studies. Scheduling work treats charging as a controllable resource: Dong et al. optimise charging-station operation under PV uncertainty and battery power constraints [12], and Tairo et al. extend this to a multi-objective microgrid energy management problem that considers fairness between users and contingency conditions alongside cost [13]. Impact studies instead take charging behaviour as given and ask what it does to the network; Kenari and Ozdemir show that correlated PV generation and fast charging jointly affect active distribution grid operation in ways that deterministic single-profile analysis does not capture [14].

The present study belongs to the second category. Charging demand is treated as an exogenous input rather than a decision variable, because the mitigation measure under investigation acts on the network rather than on charging behaviour. This is a narrower problem, but it isolates the effect being studied.

2.3 EV Charger Harmonics

Alame et al. provide the most directly relevant treatment, assessing the harmonic currents produced by EV charging on distribution systems and relating distortion to transformer stress and accelerated ageing [4]. Their central observation is that the harmonic signature depends on charger topology and loading, so a spectrum measured for one charger type cannot be assumed to represent another.

This has a direct consequence for the present work. The charger harmonic spectrum used here is the one available in the PowerFactory model rather than a measured spectrum for a specific deployed charger, and the absolute THD figures reported in Chapter 4 are conditional on it. The relative comparisons – baseline against compensated, and one converter rating against another – are much less sensitive to this, because both cases share the same source spectrum.

2.4 Power Quality Impacts and Mitigation Techniques

Harmonic mitigation approaches fall into three families. Passive filters are inexpensive and robust, but are tuned to fixed frequencies and can interact with network impedance to produce resonance. Active power filters inject compensating current under closed-loop control and adapt to changing conditions; Srivastava and Saravanan demonstrate an optimally placed active filter for EV charging stations and report substantial distortion reduction [15]. Hybrid arrangements combine both, using a passive stage for bulk filtering and a smaller active stage for the remainder [11].

A fourth option, and the one pursued here, is to use converter capability that already exists for another purpose. Ovaskainen et al. describe superposed control strategies in which a BESS provides power exchange and power-quality improvement simultaneously [8]. Behera and Ray apply the same idea to a bidirectional charging system controlling active and reactive power with harmonic compensation [9], and Pranith et al. develop a multifunctional grid-connected BESS control scheme using a second-order generalised integrator [10]. These establish feasibility. None of them, however, reports what converter rating the harmonic function required, which is precisely the question Section 4.13 addresses.

2.5 PV and Wind Integration in Microgrids

Rahman examines the effect of solar generation on the IEEE nine-bus system and identifies bus placement as a determinant of transient stability [16], and Khan et al. study coordinated stability improvement in renewable-integrated networks [17]. Kumar and Rajan argue for hybrid PV–wind architectures for EV charging on the grounds that the combination is less variable than either resource alone [5]. Srilakshmi et al. combine solar, wind, battery and EV supply through a unified power quality conditioner, optimising the design with a metaheuristic algorithm [18], which is close in spirit to the present work although it uses a dedicated conditioner rather than the storage converter.

Abd Elazim et al. are the closest single reference, addressing stability and power quality in EV charging stations supplied by hybrid energy sources through combined harmonic mitigation and load management [19].

2.6 Renewable Uncertainty and Probabilistic Analysis

Deterministic load flow answers what happens under one set of conditions. Probabilistic load flow answers what range of outcomes is possible. Mišurović and Mujović review numerical probabilistic load flow for systems with intermittent sources and note that analytical and point-estimate methods depend on linearisation or distributional assumptions that degrade in strongly nonlinear systems [6]. Constante-Flores and Illindala apply Monte Carlo simulation to a distribution system with renewable sources using a data-driven formulation [7].

Monte Carlo simulation is straightforward and makes no distributional assumption about the system response, at the cost of requiring many runs. Latin hypercube sampling reduces the number of runs needed by stratifying the input space. **Latin hypercube sampling was considered for this study but was not implemented**; the uncertainty analysis reported in Chapter 4 rests on Monte Carlo sampling alone. This is stated plainly here because earlier project documentation referred to both methods, and the distinction matters for how the results should be read.

2.7 Optimisation Methods

Tuning harmonic injection magnitude and phase is a continuous, non-convex problem in which each objective evaluation requires a full harmonic load flow. Gradients are not available, so derivative-free population methods are the natural choice.

Particle swarm optimisation moves a population of candidate solutions through the search space under the influence of each particle’s own best position and the best position found by the swarm [20]. Differential evolution generates trial vectors by adding a scaled difference between two population members to a third, then applying crossover and greedy selection [21]. Both are widely used in power-system applications and both handle the mixed magnitude–phase search here without modification, apart from one adjustment described in Section 3.15.3: phase variables are circular, so differences must be computed on the shortest arc rather than linearly.

2.8 Harmonic Standards and the Benchmark Used Here

IEEE Standard 519 provides recommended practice for harmonic control in electric power systems [22]. Its limits are not a single universal number: they depend on the voltage level, on whether voltage or current distortion is being assessed, on the short-circuit-to-load current ratio, and on the point of common coupling at which the assessment is made.

For that reason this thesis uses the phrase “**the 5 % benchmark adopted in this study**” rather than asserting that IEEE 519 requires THD below 5 % everywhere. The 5 % figure is a recognised and defensible design target for voltage distortion at distribution voltage levels, and it is applied here consistently to every bus. Establishing formal compliance would require confirming the applicable standard edition, the system voltage levels and the designated point of common coupling.

2.9 Critical Synthesis and Research Gap

Taken together, the reviewed work supports three claims and leaves one question open.

It is established that EV chargers produce network-propagating harmonic distortion [4]; that active filtering reduces it [15]; and that a storage converter can perform this function alongside its energy role [8], [9], [10].

What is not established is the capacity requirement. The active-filter sizing literature suggests the harmonic duty is small relative to other duties [11], yet this guidance derives from filters placed at the point of common coupling of the distorting load. A storage converter sited elsewhere in a meshed network must drive its compensating current through intervening impedance to reach the affected buses, and there is no reason to expect the same ratio to hold. The present study tests this directly by repeating the optimisation at two converter ratings.

2.10 Overall Methodology

The work proceeds in six stages.

1. **Network modelling.** A generic nine-bus microgrid is built in DIgSILENT PowerFactory, with PV, wind, EV charging and BESS added to the base network.
2. **Dataset preparation.** Annual hourly PV and wind profiles are taken from measured

data for Bhola, Bangladesh; the charging profile is derived from NREL EVI-Pro Lite data.

3. **Uncertainty modelling.** A 300-run Monte Carlo study perturbs renewable output using a first-order autoregressive process.
4. **Baseline harmonic study.** Harmonic load flow is run with the full system connected and BESS harmonic compensation disabled.
5. **Compensation and optimisation.** The BESS converter is represented as a harmonic current source; PSO and DE tune the injection magnitudes and phases.
6. **Sizing sensitivity.** The DE study is repeated at a reduced converter rating to identify the binding constraint.

2.11 Choice of Simulation Platform

DIgSILENT PowerFactory was selected as the simulation platform. An initial exploration phase was carried out in ETAP before the final implementation, but no ETAP result is used in this thesis. The decisive factor was PowerFactory's Python scripting interface, which allows a harmonic load flow to be executed, and bus THD values to be read back, from within an optimisation loop. Without that capability the several thousand harmonic load flows required by the optimisation would have had to be run manually.

2.12 Performance Metrics

Three metrics are used throughout Chapter 4:

- **Maximum bus THD**, which determines compliance, since the benchmark must be met at every bus.
- **System average THD**, which indicates overall network condition and distinguishes a solution that improves everywhere from one that trades buses against each other.
- **Per-bus percentage reduction** against the baseline, which shows how the benefit is distributed.

2.13 Methodological Strengths and Limitations

The principal strength of the approach is that the optimisation acts on the actual PowerFactory model rather than on a reduced analytical approximation, so network

impedance and the propagation of harmonic current are represented as the model represents them. The correlated uncertainty treatment is also more realistic than independent hourly sampling.

The limitations are equally clear. The harmonic study uses one operating point, so no annual harmonic statistic can be claimed. Compensation is confined to a single location and two harmonic orders. Latin hypercube sampling was not implemented. And, as Section 4.12 discusses, the differential evolution run was warm-started from earlier solutions, which limits what can be concluded from the comparison between the two algorithms.

2.14 Chapter Summary

Existing work establishes that EV charging causes harmonic distortion and that storage converters can mitigate it, but does not quantify the converter capacity that mitigation requires in a network where the compensator is remote from the distorting load. The methodology adopted here is designed to answer that question directly, by posing a baseline that isolates the compensation effect and by repeating the study at two converter ratings.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter documents the engineering work: the network model and how each component is represented in DIgSILENT PowerFactory, the annual dataset and its uncertainty treatment, the configuration of the harmonic study, and the two optimisation implementations that drive it.

3.2 The Generic Nine-Bus Microgrid

The network is a nine-bus microgrid built in PowerFactory. It follows the general form of the classical nine-bus test system – three synchronous generators, two transmission levels joined by transformers, and a meshed transmission section – but the load and generation placement has been modified to represent the microgrid studied here. For this reason the system is described throughout as a *generic* nine-bus microgrid rather than as the IEEE nine-bus system, since the load placement differs from the standard case and reporting it otherwise would misdescribe the model.

Table 3.1 records the placement of each element.

Table 3.1: Placement of generation, storage and load elements in the PowerFactory model.

Element	Bus	Representation
Synchronous generators	Bus 1, 2, 3	Conventional generation
Wind turbine generator	Bus 5	Static generator
PV arrays PVA1–PVA4	Bus 6	Four aggregated arrays
BESS	Bus 6	Static generator with associated harmonic source
EV charging station	Bus 4	Three charger loads
General load	Bus 8	Conventional load
Passive harmonic filter	Bus 4	RLC branch
Former DC charging station	Bus 9	Out of service

The element at Bus 9 is out of service and takes no part in any result reported in this thesis.

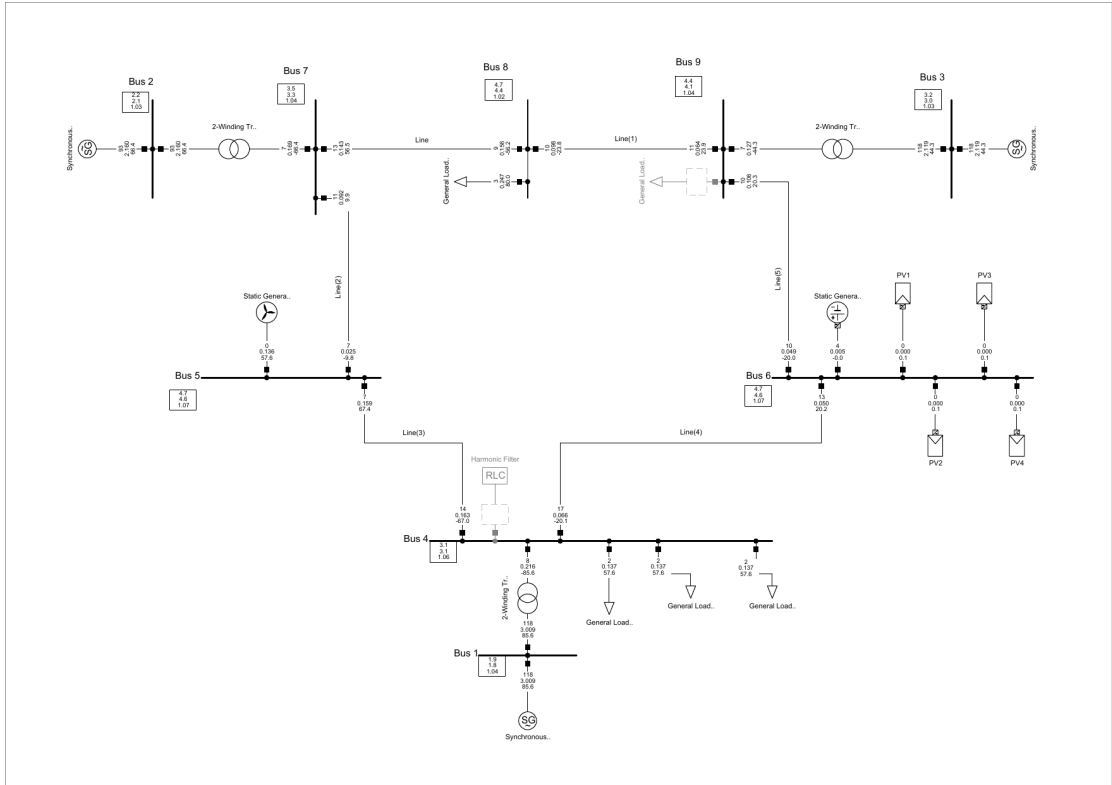


Figure 3.1: Single-line diagram of the generic nine-bus microgrid as implemented in DIgSILENT PowerFactory, showing the placement of the synchronous generators, the PV arrays, the wind turbine generator, the BESS and the EV charging station.

3.3 DIgSILENT PowerFactory Implementation

The model was implemented in PowerFactory with buses as `ElmTerm` objects named Bus 1 through Bus 9. This naming is not cosmetic: the optimisation scripts retrieve bus objects by name through `GetCalcRelevantObjects()` and read distortion from each using the `m:THD` attribute, so the naming convention is part of the interface between the model and the scripts.

3.4 Photovoltaic Model

PV generation is represented by four arrays at Bus 6, each driven by the same hourly annual profile with a peak output of 307.03 kW and an annual yield of 134,598 kWh. Total installed PV capacity is therefore 1.228 MW.

The four arrays carry identical profiles. This represents four co-located arrays experiencing the same irradiance, which is reasonable for arrays at one site, but it removes any spatial smoothing between them and makes the aggregate more variable than four dispersed arrays would be. This is a conservative choice with respect to storage sizing.

PV is modelled without harmonic current injection; Section 3.7 explains why.

3.5 Wind Model

Wind generation is represented by a single turbine at Bus 5 with a peak output of 172.08 kW and an annual yield of 43.14 MW h, giving a capacity factor of approximately 2.8 %. This is low, and it reflects the wind resource recorded in the source dataset for the site rather than a modelling choice. The practical consequence is that wind contributes relatively little annual energy but still contributes to short-term variability, which is what matters for the smoothing duty in Section 3.9.

Like PV, the wind generator is modelled without harmonic injection.

3.6 EV Charging Station Model

The EV charging station at Bus 4 is represented by three charger loads, each driven by the same hourly annual profile with a peak of 52.31 kW and an annual consumption of 127,750 kW h. The combined station peak is therefore 156.9 kW, or 0.157 MW.

Because the three chargers share one profile, their peaks are fully coincident. A real station with three chargers would show partial diversity and a lower combined peak, so the figure used here is an upper bound on station demand. This too is conservative for sizing purposes.

3.7 EV Charger Harmonic Source

The charging station is the only harmonic source in the model. Its spectrum, as configured in PowerFactory, is given in Table 3.2.

Table 3.2: Harmonic current spectrum configured for the EV charger, expressed as a percentage of the reference current.

Harmonic order	Magnitude (%)	Used in compensation
3rd	2.8352	No
5th	2.6949	Yes
7th	1.5853	Yes
9th	1.3670	No

Charger rated current ≈ 0.2510 kA.

Compensation targets the fifth and seventh orders. These are the dominant non-triplen

orders in the spectrum and the ones that propagate most readily through the network; triplen harmonics such as the third and ninth are largely confined by the transformer winding configurations and are less amenable to compensation by a source located elsewhere in the network.

3.7.1 PV, Wind and Normal BESS Operation Inject No Harmonics

PV, wind and the BESS in its energy-storage role are modelled purely as fundamental-frequency sources. In reality all three are inverter-interfaced and produce some distortion, so this is a modelling simplification.

It has an important consequence for how the results must be read. Adding PV, wind or a conventionally operated BESS to the model cannot change bus THD by any mechanism other than altering the fundamental power flow and hence the fundamental voltage. **Any observation that DER integration leaves distortion largely unchanged is therefore a consequence of the model, not an empirical finding about DER.** This is stated explicitly here so that no such claim is made in Chapter 4.

3.8 BESS Model and Operating Concept

The BESS is located at Bus 6 and performs two distinct functions that must be kept separate.

3.8.1 Energy Storage Function

At the fundamental frequency, the converter exchanges real power with the network: absorbing power to charge and delivering power to discharge, smoothing renewable variability and supporting the charging station peak.

3.8.2 Active Harmonic Compensation Function

The converter is also a controllable current source and can be commanded to produce current at harmonic frequencies. Its total current is the superposition

$$i_{\text{BESS}} = i_1 + i_5 + i_7 + \dots \quad (3.1)$$

where i_1 is the fundamental component associated with charging or discharging, and i_5 , i_7 are the controlled compensating components.

It is worth stating what this does and does not mean. The battery is not charging and discharging simultaneously; that would be meaningless. The converter exchanges real power in one direction at the fundamental frequency while, in the same instant, shaping current components at 250 Hz and 350 Hz. Those components carry no net real power over a cycle. The converter can therefore compensate while charging, while discharging, or at approximately zero net power, provided the DC link remains energised.

3.8.3 Converter Current Headroom

The functions are not independent, because they share the same converter current capability. The components combine in an RMS sense:

$$I_{\text{RMS}}^2 \approx I_1^2 + I_5^2 + I_7^2 + \dots \leq I_{\text{rated}}^2 \quad (3.2)$$

The harmonic current available for compensation is therefore whatever remains after the fundamental duty is met. Heavy charging or discharging reduces the headroom for compensation, and a converter sized only for its energy duty may have very little to spare. This relationship is the physical basis for the sizing result reported in Section 4.13.

3.9 BESS Sizing

Sizing proceeds from the system's own profiles and from published practice for each duty. Table 3.3 sets out the derivation.

3.9.1 Renewable Smoothing Duty

Installed renewable capacity is 1.400 MW. Reported BESS-to-PV power ratios span a wide range: the LG&E/KU E.W. Brown facility pairs a 10 MW PV plant with a 1 MW BESS at two-hour duration [23]; the Karratha Airport hybrid station sized its inverter at 47 % of each PV block's nominal AC power to meet a ramp-rate requirement [24]; and a ramp-rate control study found 60 % of PV nominal power adequate to smooth nearly all detected ramps [25]. Practical designs cluster around 20 to 30 %. Adopting 25 % gives 0.35 MW.

The 25 % figure is an engineering assumption drawn from the middle of the observed range, not a standard requirement, and is treated as such.

Table 3.3: Derivation of the BESS power and energy rating from the system’s own profiles and from published sizing practice.

Basis	Calculation	Result
Installed renewable capacity	$4 \times 307.03 \text{ kW PV} + 172.08 \text{ kW wind}$	1.400 MW
EV charging station peak	$3 \times 52.31 \text{ kW}$	0.157 MW
Power rating: smoothing duty	25% of 1.400 MW	$\approx 0.35 \text{ MW}$
Power rating: EV peak shaving	$\approx 100\%$ of 0.157 MW	$\approx 0.16 \text{ MW}$
Power rating: harmonic headroom	Small margin on the larger of the above	(see Section 4.13)
Derived power rating	Larger duty plus harmonic margin	$\approx 0.4 \text{ MVA}$
Energy capacity	2-hour duration at 0.4 MW	$\approx 0.8 \text{ MW h}$
Rating adopted in simulation	Set by harmonic performance, not by the derivation above	1.5 MVA / 0.8 MW h

3.9.2 EV Peak-Shaving Duty

Sizing for peak shaving is framed around the station’s own peak demand. Koolman et al. size BESS and grid-tie capacity for an EV fast-charging hub using a multi-objective formulation against real demand data [26]. Taking approximately full coverage of the station peak gives 0.16 MW.

3.9.3 Harmonic Compensation Duty

The published guidance here is striking. A hybrid active/passive filter design achieves compliant distortion using active inverters rated at 2.2 % and 0.73 % of the 20 MVA load being compensated [11]. On this basis harmonic duty would require only a small margin on top of the other duties and would not drive the rating at all.

3.9.4 Energy Capacity

NREL’s assessment of deployed grid-scale storage finds that roughly 90 % of installed capacity has a duration of four hours or less [27], and NREL’s Annual Technology Baseline uses four hours as its default utility-scale configuration [28]. A two-hour duration at 0.4 MW gives 0.8 MW h, consistent with the two-hour LG&E/KU installation [23].

3.9.5 The Rating Actually Adopted

The derivation above yields approximately 0.4 MVA. The rating adopted in the simulation is 1.5 MVA with 0.8 MW h of energy capacity.

This is a large discrepancy and it is not an error. The 1.5 MVA rating was not derived from the duties above; it was established empirically, because smaller ratings did not achieve the distortion target. Section 4.13 presents that evidence and examines why the harmonic duty in this network is so much more demanding than the published active-filter guidance would predict. The derivation is retained here because the comparison between the two is itself a result.

3.10 Annual Input Datasets

Table 3.4: Annual input profiles used in the PowerFactory model. All series are hourly over a full year (8760 points).

Element	Bus	Peak (kW)	Annual (kWh)	Note
PVA1–PVA4	Bus 6	307.03	134,598	Four identical arrays Capacity factor $\approx 2.8\%$
WTG1	Bus 5	172.08	43,144	
Charger 1–3	Bus 4	52.31	127,750	Three identical profiles

PV and wind profiles derive from the KU-HMG2 hybrid microgrid dataset for Bhola, Bangladesh [29], which provides hourly measurements over a full year. The charging profile derives from NREL EVI-Pro Lite data [30], which models charging demand from vehicle usage patterns rather than assuming a shape. All series are hourly over 8760 points.

Using measured rather than synthetic generation data means the profiles carry the actual intermittency of the site, including extended low-output periods that a smooth synthetic profile would not reproduce.

3.11 Scenario Generation and Uncertainty Implementation

A Monte Carlo study of 300 runs was performed. In each run the PV and wind series were perturbed by a multiplicative noise process with first-order autoregressive structure, so that the perturbation at hour t depends on the perturbation at hour $t - 1$:

$$\varepsilon_t = \phi \varepsilon_{t-1} + \eta_t \quad (3.3)$$

where η_t is an independent random innovation and ϕ is the autoregressive coefficient.

The reason for using an autoregressive rather than an independent perturbation is physical. Cloud cover and wind conditions persist across hours. Independent hourly sampling produces a series that oscillates implausibly between high and low output and, more importantly, understates the probability of sustained low-generation periods – exactly the conditions that determine storage requirements.

Across the 300 runs, PV peak output ranged from 174.5 kW to 533.4 kW and wind peak from 100.0 kW to 858.7 kW. EV charging demand was held effectively constant across runs, with a peak standard deviation of 0.004 kW. The uncertainty analysis in this study is therefore a study of *renewable* uncertainty; charging demand uncertainty was not varied, and Section 4.20 records this as a limitation.

As a validation check, PV output at night was confirmed to be zero in all 300 runs (maximum night-time PV = 0 kW), confirming that the multiplicative perturbation did not introduce generation at hours when the underlying series was zero.

Five scenarios were exported for detailed examination alongside the deterministic base case; their statistics appear in Table 4.1.

3.11.1 Relationship Between the Scenario Study and the Harmonic Study

The scenario study and the harmonic study operate at different levels and are not combined. The scenario study characterises the range of operating conditions the network may experience over a year. The harmonic study is executed at a single operating point. No harmonic result in this thesis is an annual average, and none is presented as one. Section 5.8 sets out what a genuine annual harmonic study would require.

3.12 Harmonic Load Flow Configuration

Harmonic analysis uses the PowerFactory harmonic load flow command (ComHlfd) with the all-frequencies option enabled (`iopt_allfrq = 1`), so that all configured harmonic orders are evaluated in one execution. After each execution the voltage THD at each of the nine buses is read from the `m:THD` attribute of the corresponding terminal object.

A single execution takes a fraction of a second, which is what makes the optimisation

feasible: the differential evolution run performs several thousand such executions.

3.13 Definition of Study Cases

Two comparisons are possible, and they answer different questions. Distinguishing them is essential.

Table 3.5: Definition of the study cases and the question each addresses.

Comparison	Cases	Question answered
DER integration	No DER versus PV + wind + BESS + EV	What does adding distributed generation do to the network?
Harmonic mitigation (used in this thesis)	PV + wind + EV + BESS, compensation off versus the same system, compensation on	What does BESS harmonic compensation achieve?

The second comparison is the one used throughout Chapter 4. The baseline is the complete system – PV, wind, EV charging station and BESS all in service – with the BESS harmonic compensation disabled. In the scripts this is realised by setting all harmonic injection magnitudes to zero (`hsrc.kba1 = [0, 0, 0]`) and running one harmonic load flow before the search begins.

This construction is what makes the mitigation claim valid. Baseline and compensated cases differ in exactly one respect: whether the converter is injecting compensating current. Any change in THD between them is attributable to compensation and to nothing else.

3.14 BESS as an Active Harmonic Compensator

The compensating function is represented by a balanced, phase-correct harmonic current source associated with the BESS, with harmonic currents referred to the converter’s rated current. The phase-correct representation is essential: a spectrum specified by magnitude alone, as in the standard IEC source types, carries no phase information, and phase is precisely the quantity that determines whether injected current cancels or reinforces the existing distortion.

At each harmonic order, the net current at the affected bus is the phasor sum

$$\bar{I}_{net,h} = \bar{I}_{charger,h} + \bar{I}_{BESS,h} \quad (3.4)$$

Cancellation is most effective when $\bar{I}_{BESS,h}$ approaches the negative of $\bar{I}_{charger,h}$ as seen at that bus. Because the two sources are at different buses and the network between them has frequency-dependent impedance, the ideal phase is not simply 180° : it must be found for the network as built. This is why the optimisation searches phase as well as magnitude.

3.15 Optimisation Formulation

3.15.1 Decision Variables

The PSO implementation searches six variables – magnitude and phase of the fifth, seventh and eleventh harmonics. The DE implementation searches four, fixing the eleventh harmonic at zero. This reduction followed from the PSO results, in which the eleventh-harmonic magnitude converged to zero: the search was assigning it no useful role, and removing it reduced the dimensionality of the problem.

Magnitudes are bounded to 0 to 100 % of rated current. Phases are bounded to 0 to 360° and wrapped modulo 360.

3.15.2 Objective Functions

The two algorithms use different objectives, and the difference is consequential.

PSO minimises the worst-case normalised distortion ratio across the nine buses, with the system mean as a tie-breaker:

$$\min \max_{i \in \{1..9\}} \frac{\text{THD}_i^{comp}}{\text{THD}_i^{base}}, \quad \text{tie-break} \min \frac{1}{9} \sum_{i=1}^9 \frac{\text{THD}_i^{comp}}{\text{THD}_i^{base}} \quad (3.5)$$

This rewards proportional improvement at every bus. Its weakness is that it is indifferent to absolute compliance: a bus can improve by a large proportion and still exceed the benchmark.

DE therefore uses a feasibility-first lexicographic objective. For a target $T = 4.99\%$ and excess $e_i = \max(0, \text{THD}_i - T)$, candidates are ranked on the ordered tuple

$$\left(\max_i e_i, \sum_i \mathbb{I}[e_i > 0], \sum_i e_i^2, \max_i \text{THD}_i, \frac{1}{9} \sum_i \text{THD}_i \right) \quad (3.6)$$

compared element by element. The first criterion drives the worst violation towards zero; once no bus violates the target, the first three terms vanish and the search continues on maximum and then average distortion. This directly encodes the requirement that *every* bus must comply, which Eq. (3.5) does not.

3.15.3 Constraint Handling

Magnitudes are clipped to their bounds after each update. Phases are wrapped modulo 360° , and – importantly – all phase differences are computed on the shortest arc:

$$\Delta\theta = ((\theta_a - \theta_b + 180) \bmod 360) - 180 \quad (3.7)$$

Without this, the difference between 350° and 10° would be evaluated as 340° rather than -20° , and both algorithms would be driven away from optima lying near the wrap point. Since the optimised phases in Chapter 4 lie between 292° and 335° , this is not a theoretical concern.

3.16 Particle Swarm Optimisation Implementation

Each particle’s position is a candidate harmonic configuration. Velocity is updated from inertia, the particle’s own best position and the swarm’s best:

$$v_d^{k+1} = w v_d^k + c_1 r_1 (p_d^{best} - x_d^k) + c_2 r_2 (g_d^{best} - x_d^k) \quad (3.8)$$

with r_1, r_2 drawn uniformly on $[0, 1]$. Inertia decreases linearly from 0.9 to 0.35 across the run, shifting the swarm from exploration to refinement. Velocities are clamped to 25 % of each dimension’s range. Settings are listed in Table 3.6; a pattern search around the final swarm best follows the main loop.

3.17 Differential Evolution Implementation

For each individual $\mathbf{x}_i$, three distinct others are selected at random and a mutant formed:

$$\mathbf{v} = \mathbf{x}_{r_1} + F (\mathbf{x}_{r_2} - \mathbf{x}_{r_3}) \quad (3.9)$$

with F drawn from $\mathcal{U}(0.45, 0.90)$ for each trial rather than held fixed, which varies the mutation step and reduces the chance of stagnation. Phase components of the difference use Eq. (3.7). Binomial crossover with $CR = 0.90$ combines mutant and parent, with one dimension taken from the mutant unconditionally so that the trial always differs from its parent. Selection is greedy under Eq. (3.6). A five-level coordinate polish, with step sizes decreasing from 5 to 0.25, refines the final solution.

Table 3.6: Settings used for the two optimisation algorithms, as implemented in the PowerFactory Python scripts reproduced in Appendix A.

Setting	PSO	Differential Evolution
Decision variables	6 (magnitude and phase of H5, H7, H11)	4 (magnitude and phase of H5, H7; H11 fixed at zero)
Population size	24 particles	60 individuals
Iterations	45	100 maximum; terminated at generation 8
Control parameters	$w: 0.9 \rightarrow 0.35$, $c_1 = 1.6$, $c_2 = 1.8$, velocity clamp 25% of range	$CR = 0.90$, $F \sim \mathcal{U}(0.45, 0.90)$
Magnitude bounds	0–100% of rated current	0–100% of rated current
Phase bounds	0–360°, wrapped	0–360°, wrapped
Objective	Minimise worst-case THD ratio against baseline; tie-break on mean ratio	Lexicographic feasibility-first towards a 4.99% target
Initialisation	1 seeded particle, 23 random	5 seeded, 15 perturbed, 40 random
Local refinement	Pattern search, 3 rounds	Coordinate search, 5 step levels
Random seed	42	42

3.17.1 Population Initialisation

The DE population was not initialised randomly. Five individuals were seeded with solutions from earlier optimisation stages, including the PSO result; fifteen were random perturbations of those seeds; the remaining forty were uniformly random.

This warm start was intended to make the search converge quickly, and it did. It also has a consequence that must be acknowledged rather than glossed over: because the seeded population already contained good solutions, the run cannot be read as evidence that DE

independently discovers better solutions than PSO. Section 4.12 treats this directly.

3.18 Simulation Workflow

1. Build the network in PowerFactory and assign annual profiles to PV, wind and the chargers.
2. Generate uncertainty scenarios by Monte Carlo sampling (Section 3.11).
3. Set BESS harmonic injection to zero, run harmonic load flow, and record the baseline THD at all nine buses.
4. Run the optimisation: apply a candidate harmonic configuration, execute harmonic load flow, read THD at all nine buses, evaluate the objective, update the population, and repeat.
5. Apply local refinement to the best solution found.
6. Re-apply the final configuration and execute a confirming harmonic load flow.
7. Repeat steps 3–6 at a reduced converter rating for the sizing sensitivity study.

3.19 Verification

Three checks were applied. The final reported configuration is re-applied and a confirming harmonic load flow executed, so that the reported THD values come from a fresh evaluation rather than from stored search state. Both scripts use a fixed random seed (42), so each run is exactly reproducible. And the baseline is re-derived within each script rather than carried over between runs, ensuring that the reported reductions are computed against the correct reference.

3.20 Engineering Feasibility and Economic Considerations

Implementing this scheme requires no additional plant beyond the BESS, but does require a converter control capability able to superimpose commanded harmonic current on the fundamental reference, and sufficient current headroom for it.

On availability: BESS units in the derived capacity range are manufactured within Bangladesh. Fakir Technologies Ltd. produces the ZERO product line, with a mid tier spanning 100 kW to 800 kW that brackets the 0.4 MVA to 0.5 MVA figure from Section 3.9, and has deployed a 1 MW industrial unit [31]. Imported containerised

systems are also available in this range. The 1.5 MVA rating the harmonic duty actually requires sits above the domestic mid tier, which is itself part of the cost consequence discussed in Section 4.16.

3.21 Complex Engineering Problem Attributes

The work satisfies attribute P1 through the depth of knowledge required across power system analysis, power quality and optimisation (knowledge profiles K4, K6 and K8). It satisfies P2 through genuinely conflicting requirements: converter rating against cost, harmonic headroom against fundamental power capability, and compensation at one bus against distortion at another. It satisfies P4 in that coordinated renewable, storage and EV operation under uncertainty is not a routine analysis, and P7 in that charging behaviour, renewable generation, storage operation, power flow and harmonic propagation are coupled and cannot be treated separately.

3.22 Chapter Summary

The model, the dataset, the uncertainty treatment, the harmonic study configuration and both optimisation implementations have been documented. The baseline has been defined so that the mitigation comparison isolates the effect of compensation. Chapter 4 presents what the model produced.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents the results and interprets them. For each result four questions are addressed: what happened, why it happened, whether it meets the research objective, and what limitation remains.

4.2 Base System Operation

4.3 Renewable and Load Profile Analysis

The annual profiles establish the operating envelope within which the harmonic study sits.

PV generation peaks at 307.03 kW per array, four arrays giving 1.228 MW installed, but with an annual yield per array of only 134.6 MW h the capacity factor is modest and output is zero for roughly half of all hours. Wind contributes 43.14 MW h annually at a capacity factor near 2.8 %, so its role is short-term variability rather than bulk energy.

Against this, the EV charging station draws 127.75 MW h per charger annually with a coincident station peak of 156.9 kW.

Two features of the resulting net load matter. Renewable penetration across scenarios lies between 15.97 % and 17.38 % (Table 4.1), so the network remains substantially dependent on conventional generation. And net load becomes negative at some hours, reaching -118.6 kW in the deterministic case: at those hours renewable generation exceeds total demand and power flows in reverse. Even at a modest average penetration, therefore, the network already experiences reverse flow conditions – a point relevant to protection coordination that this study does not otherwise pursue.

Table 4.1: Annual energy and net-load statistics for the deterministic base case and the five exported Monte Carlo scenarios. Source: scenario generation script output.

Scenario	PV (MWh)	Wind (MWh)	Total load (MWh)	Net peak (kW)	RE pen. (%)
Base (deterministic)	141.55	47.49	1183.96	328.22	15.97
Scenario 1	137.35	59.01	1185.14	343.77	16.57
Scenario 2	140.52	63.47	1173.44	341.13	17.38
Scenario 3	140.90	59.39	1185.57	340.48	16.89
Scenario 4	142.75	55.88	1169.17	339.92	16.99
Scenario 5	137.25	58.70	1176.16	345.97	16.66

4.4 Uncertainty and Scenario Results

Across 300 Monte Carlo runs, PV peak output ranged from 174.5 kW to 533.4 kW and wind peak from 100.0 kW to 858.7 kW. The wind range is the wider in relative terms, consistent with the greater short-term volatility of the resource.

Two observations about what this does and does not show.

First, the effect on annual energy is far smaller than the effect on peaks. Total load energy across the five exported scenarios varies only between 1169.2 MWh and 1185.6 MWh, a spread of about 1.4 %, while net-load peak varies over 328.2 kW to 346.0 kW. Uncertainty expresses itself in extremes rather than in totals, which is the case for sizing storage against peak conditions rather than average ones.

Second, EV charging demand was effectively identical across runs, with a peak standard deviation of 0.004 kW. This is a study of renewable uncertainty, not of combined renewable and demand uncertainty, and the conclusions should be read accordingly.

The validation check that PV output remained zero at night in all 300 runs (maximum night-time PV = 0 kW) confirms the perturbation did not introduce generation where the underlying series had none.

4.5 Baseline Harmonic Distortion

With all elements in service and BESS harmonic compensation disabled, harmonic load flow gives the distortion in Table 4.2.

Table 4.2: Baseline voltage total harmonic distortion at all nine buses. Full system connected (PV, wind, EV charging station and BESS in service) with BESS harmonic compensation disabled. Source: PowerFactory harmonic load flow.

Bus	Voltage THD (%)	Status vs. 5% benchmark
Bus 1	3.3277	Compliant
Bus 2	3.0019	Compliant
Bus 3	5.0410	Non-compliant
Bus 4	5.6176	Non-compliant
Bus 5	6.2567	Non-compliant
Bus 6	5.6363	Non-compliant
Bus 7	4.7471	Compliant
Bus 8	8.0781	Non-compliant (worst)
Bus 9	6.8270	Non-compliant
Maximum	8.0781	at Bus 8
Average	5.3926	5 of 9 buses non-compliant

4.5.1 Observed Distortion Pattern

Voltage THD reaches 8.0781 % at Bus 8. The system average is 5.3926 %, and five of the nine buses – buses 3, 4, 5, 6, 8 and 9 – meet or exceed the 5 % benchmark. Only buses 1, 2 and 7 comply.

4.5.2 Cause of the Distortion Pattern

The charging station at Bus 4 is the only harmonic source. Its injected current flows outward through the network, and the voltage distortion produced at any bus depends on the harmonic impedance seen between that bus and the source. Buses electrically remote from strong generation exhibit higher distortion because there is less low-impedance support at harmonic frequencies to hold the waveform.

Bus 8 is the worst affected despite not being the injection bus. This is characteristic of harmonic propagation: distortion is a network-level phenomenon, and the bus with the highest distortion is often not the one hosting the source. It is a reason why mitigation targeted only at the charging station bus would be insufficient here, and why the objective functions in Section 3.15.2 are system-wide.

4.5.3 Against the Objective

This satisfies research objective 3 and answers RQ1. Bus 8 is the critical bus; Bus 5 at 6.2567 % and Bus 9 at 6.8270 % are next.

4.5.4 Limitation

These values correspond to one operating condition. They are not annual averages, and no annual harmonic statistic is claimed. They are also conditional on the charger harmonic spectrum configured in the model (Table 3.2), which is a representative rather than a measured spectrum for a specific deployed charger.

4.6 Critical Buses

Ranking by baseline distortion gives Bus 8 (8.0781 %), Bus 9 (6.8270 %), Bus 5 (6.2567 %), Bus 6 (5.6363 %) and Bus 4 (5.6176 %). Any mitigation scheme must bring Bus 8 down by at least 38.1 % simply to reach the benchmark, which sets the difficulty of the problem.

4.7 Effect of DER Integration

No result is presented for this comparison, and the reason should be stated plainly.

As established in Section 3.7, PV, wind and the BESS in its energy role are modelled without harmonic injection. Adding them to the model cannot alter bus THD except through changes in the fundamental voltage. Any finding that DER integration leaves distortion unchanged would therefore be a restatement of the modelling assumption, not evidence about DER behaviour.

Reporting such a comparison as a result would be misleading, so it is omitted. Section 5.8 notes what would be required to address the question properly.

4.8 Harmonic Mitigation Using Particle Swarm Optimisation

PSO reduced the worst-case ratio from 0.8371 to 0.8043, a worst-case reduction of 19.59 %, with an average ratio of 0.6778 corresponding to 32.22 % average reduction. The optimised settings were H5 at 90.279 % magnitude and 334.9° phase and H7 at 98.714 % and 320.5°; the eleventh-harmonic magnitude converged to zero.

4.8.1 PSO Reduced Distortion but Left Bus 5 Non-Compliant

Distortion fell at every bus. But Bus 5 settled at 5.0285 %, marginally above the benchmark, and the run therefore failed the compliance requirement despite improving

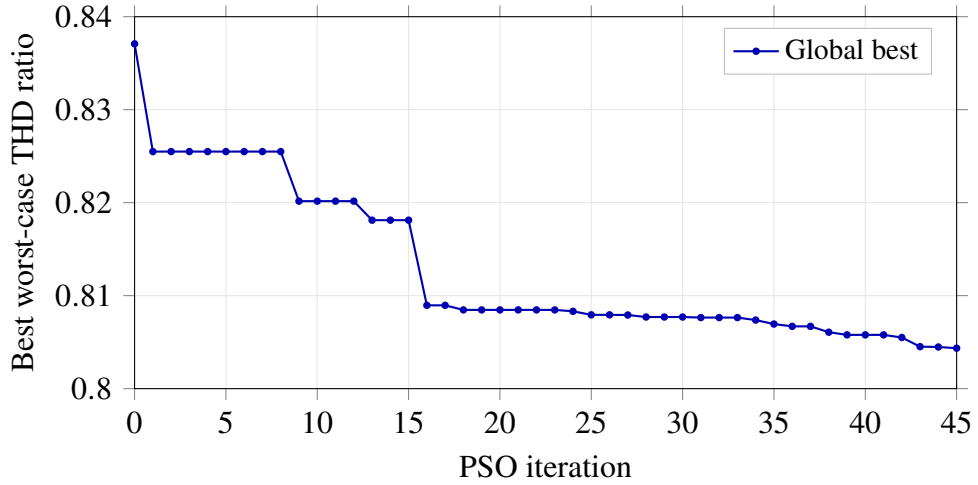


Figure 4.1: PSO convergence: best worst-case THD ratio against baseline over 45 iterations. A ratio of 1.0 corresponds to no improvement.

the whole network.

The reason is the objective function. Equation (3.5) minimises the worst *ratio* – proportional improvement – and is indifferent to absolute values. A bus already low in the baseline that improves only slightly can dominate the objective, while a bus that improves substantially but remains just above 5 % attracts no special penalty. The search was optimising the wrong thing for the requirement in hand, and this motivated the reformulated objective used for DE.

The convergence trace in Fig. 4.1 shows long plateaux with small step improvements, ending at 0.8043 with the curve still descending slowly. The swarm was not exhausted; it was making progress too slowly to matter within the iteration budget.

4.9 Harmonic Mitigation Using Differential Evolution

DE reduced the maximum bus distortion to 4.6866 % at Bus 6 and the system average to 3.5907 %, bringing all nine buses below the benchmark. The optimised settings were H5 at 100 % magnitude and 292.83° phase and H7 at 100 % and 324.97°, with H11 fixed at zero.

Relative to the baseline, maximum distortion fell by 42.0 % and system average by 33.4 %.

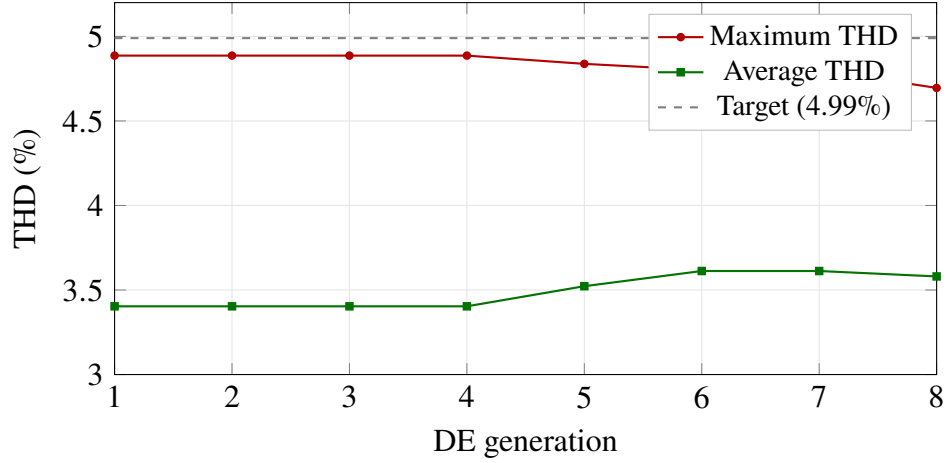


Figure 4.2: Differential evolution convergence over the eight generations executed before the termination criterion was met. Note that the maximum THD of the best individual is already below the target at generation 1, a consequence of the seeded initial population discussed in Section 4.12.

4.9.1 The Feasibility-First Objective Explains DE's Success

The lexicographic objective of Eq. (3.6) treats a bus above the target as a constraint violation rather than as a poor ratio. Every candidate with a violation ranks below every candidate without one, regardless of how good the violating candidate looks on average. The search is therefore driven to eliminate violations first and improve distortion second. This is precisely the structure the compliance requirement demands, and it is the substantive difference between the two runs.

4.10 Bus-Wise Comparison

4.10.1 An Observation About Transformer-Coupled Bus Pairs

Table 4.3 shows an unmistakable pattern. Buses 1 and 4 record identical reductions (44.38 % under DE, 40.40 % under PSO); so do buses 2 and 7 (26.43 %, 31.46 %) and buses 3 and 9 (36.06 %, 32.54 %). The same pairing appears in the reduced- rating case in Table 4.4.

These pairs are exactly the bus pairs joined by a two-winding transformer in the network (Fig. 3.1). The proposed explanation is that the harmonic voltage ratio across each transformer is essentially fixed by the transformer turns ratio and impedance, so proportional changes in distortion on one side appear undiminished on the other. If correct, this is a useful structural insight: mitigation effort directed at one bus of such a pair benefits its partner automatically, and the nine buses effectively present six

Table 4.3: Bus-wise voltage THD before and after BESS harmonic compensation, for both optimisation algorithms at the 1.5 MVA converter rating. Source: PowerFactory harmonic load flow driven by the PSO and DE scripts.

Bus	Baseline	PSO		DE	
	THD (%)	THD (%)	Red. (%)	THD (%)	Red. (%)
Bus 1	3.3277	1.9834	40.40	1.8507	44.38
Bus 2	3.0019	2.0576	31.46	2.2085	26.43
Bus 3	5.0410	3.4007	32.54	3.2230	36.06
Bus 4	5.6176	3.3482	40.40	3.1243	44.38
Bus 5	6.2567	5.0285	19.63	4.6860	25.10
Bus 6	5.6363	4.5322	19.59	4.6866	16.85
Bus 7	4.7471	3.2538	31.46	3.4925	26.43
Bus 8	8.0781	4.6851	42.00	4.6800	42.07
Bus 9	6.8270	4.6056	32.54	4.3649	36.06
Maximum	8.0781	5.0285	–	4.6866	–
Average	5.3926	3.6550	–	3.5907	–

Buses that remain at or above the 5 % benchmark are Bus 5 under PSO (5.0285%); none under DE.

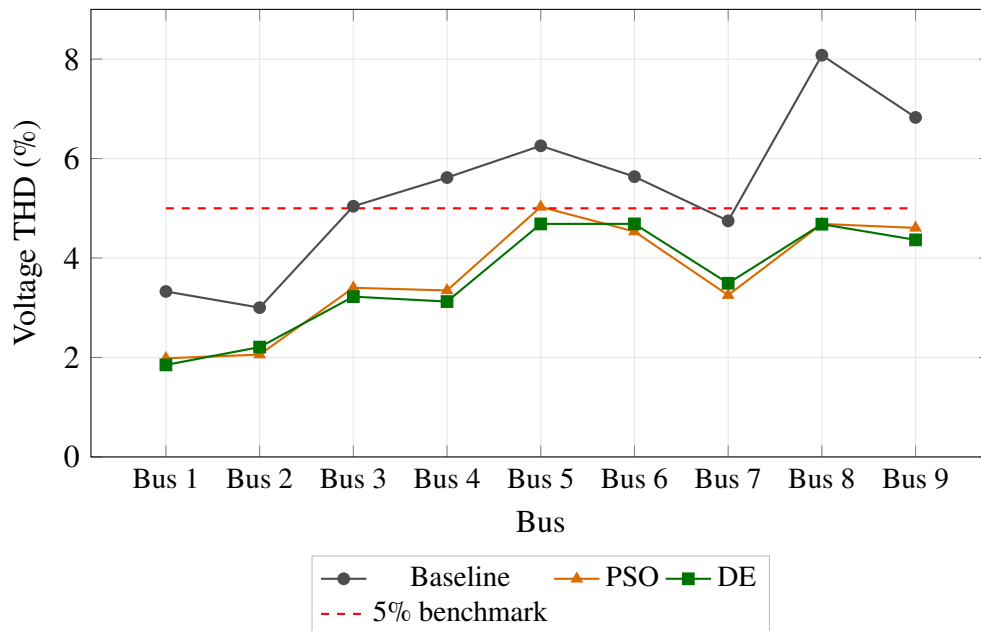


Figure 4.3: Voltage THD at each bus for the baseline and for both optimisation algorithms at the 1.5 MVA converter rating. Only the DE result lies below the benchmark at every bus.

independent targets rather than nine.

4.11 Harmonic Magnitude and Phase

The optimised phases are informative. DE selected 292.83° for the fifth harmonic and 324.97° for the seventh; PSO selected 334.9° and 320.5° . None is close to 180° .

This confirms the argument made in Section 3.14. If the compensator and the distorting load shared a bus, direct opposition at 180° would be close to optimal. They do not: the charging station is at Bus 4 and the BESS at Bus 6, and the compensating current must traverse network impedance that is frequency-dependent and introduces its own phase shift. The optimal injection phase compensates for that traversal, and it differs between the fifth and seventh harmonics because the network impedance differs at 250 Hz and 350 Hz.

The magnitudes are more troubling. DE selected 100 % for both orders – exactly the imposed upper bound. This is discussed in Section 4.13 and is the most important result in this chapter.

4.12 Algorithm Comparison

DE achieved a lower maximum distortion than PSO (4.6866 % against 5.0285 %) and a marginally lower average (3.5907 % against 3.6550 %), and satisfied the benchmark where PSO did not.

It does not follow that differential evolution is the better algorithm for this problem, and the thesis does not claim so. Three facts prevent that conclusion.

First, the two runs used different objective functions. DE's advantage on compliance is attributable to the feasibility-first formulation of Eq. (3.6), which explicitly targets absolute compliance, rather than to the search mechanism. PSO given the same objective might well have found a compliant solution.

Second, the DE population was seeded with solutions from earlier stages, including the PSO result (Section 3.17.1). Figure 4.2 shows the best individual already at 4.8865 % at generation 1, below the 4.99 % target. A feasible solution was present in the initial population before any evolutionary operator had acted. The run terminated at generation 8 when the success counter reached its threshold. DE therefore refined an existing good solution rather than discovering one.

Third, both runs used a single random seed. No repeatability assessment was carried out, so the difference between two single runs cannot be separated from run-to-run variation.

The defensible statement is narrower and more useful: *a feasibility-first objective combined with a warm-started differential evolution search produced a compliant solution where a ratio-based particle swarm search from a largely random population did not*. Establishing an algorithmic ranking would require unseeded runs of both algorithms under the same objective across multiple seeds. This is recorded in Section 5.8.

4.13 Converter Sizing and the Harmonic Capacity Limitation

This section presents what the authors consider the principal finding.

4.13.1 The Saturation Signal

In the successful DE result, both harmonic magnitudes converged to exactly 100 % of rated current – the imposed upper bound. An optimiser that stops at a bound is not reporting an interior optimum; it is reporting that it would have used more of the resource had more been available. The achieved distortion of 4.6866 % is thus the best attainable with the available harmonic current, not the best attainable in principle.

4.13.2 The Reduced-Rating Test

To test this directly, the DE study was repeated with a 1.0 MVA converter, all other settings unchanged.

The result was unambiguous. Maximum distortion rose to 5.6729 % at Bus 8, and Bus 6 at 5.1132 % also exceeded the benchmark. The search reported that no feasible solution existed within the magnitude bounds. Both magnitudes again saturated at 100 %, at 347.56° and 333.07°.

4.13.3 Interpretation

The two cases differ only in converter rating. The algorithm, its settings, the bounds, the network and the operating point are identical. The transition from non-compliance to compliance is therefore attributable to converter rating alone.

This answers RQ4 directly: **the limit on achievable harmonic mitigation in this system is the available converter current, not the optimisation procedure**. Running more

Table 4.4: Effect of BESS converter rating on achievable harmonic mitigation. Both cases use differential evolution with identical settings and identical harmonic magnitude bounds. Source: PowerFactory harmonic load flow.

Bus	1.0 MVA converter		1.5 MVA converter	
	THD (%)	Red. (%)	THD (%)	Red. (%)
Bus 1	2.1616	35.04	1.8507	44.38
Bus 2	2.4594	18.07	2.2085	26.43
Bus 3	3.3405	33.73	3.2230	36.06
Bus 4	3.6491	35.04	3.1243	44.38
Bus 5	3.9875	36.27	4.6860	25.10
Bus 6	5.1132	9.28	4.6866	16.85
Bus 7	3.8892	18.07	3.4925	26.43
Bus 8	5.6729	29.77	4.6800	42.07
Bus 9	4.5241	33.73	4.3649	36.06
Maximum	5.6729	—	4.6866	—
Average	3.8664	—	3.5907	—
Outcome: 1.0 MVA — Bus 6 and Bus 8 remain non-compliant. 1.5 MVA — all nine buses compliant.				

generations, adding particles or changing algorithm would not have rescued the 1.0 MVA case, because the binding constraint was physical rather than computational.

It also explains the 1.5 MVA rating adopted in Section 3.9, which the duty-based derivation could not justify. The rating was set by harmonic performance. From the sizing derivation, roughly 0.4 MVA serves the smoothing and peak-shaving duties, leaving approximately 1.1 MVA as headroom for harmonic compensation.

4.13.4 Disagreement With Published Sizing Practice

This finding sits in direct tension with the active-filter sizing literature. The hybrid filter design cited in Section 3.9 achieves compliant distortion using active inverters rated at 2.2 % and 0.73 % of the compensated load [11]. In the present system the harmonic duty required roughly 275 % of the rating needed for all other duties combined. The disparity is more than two orders of magnitude in the ratio.

The thesis does not claim the published guidance is wrong. It proposes that the two situations are not comparable, for reasons that follow from the model:

1. **Location.** A conventional active filter is installed at the point of common coupling of the distorting load, where it can cancel harmonic current close to source. The BESS here is at Bus 6 while the charging station is at Bus 4. Compensating current must

traverse intervening network impedance, and only part of it reaches the buses where distortion is worst.

2. **Objective.** A filter at the PCC is required to clean one connection point. The compensator here is required to bring *all nine buses* below the benchmark simultaneously, with Bus 8 – the worst bus – neither the injection point nor adjacent to it.
3. **Reference base.** Magnitudes here are percentages of the converter’s own rated current, whereas the cited figures are percentages of the compensated load. The two denominators differ, so the ratios are not directly comparable even before the physical differences are considered.

The practical implication is that published active-filter sizing ratios should not be transferred to a storage converter sited remotely from the distorting load without checking. Where a BESS is intended to serve a harmonic function in addition to its energy function, that function may dominate the converter rating.

4.13.5 Limitation

This conclusion rests on two converter ratings. A finer sweep – 1.1, 1.2, 1.3, 1.4 MVA – would locate the threshold and permit a mitigation-versus-rating curve rather than two points. Section 5.8 records this.

4.14 Evaluation Against the Benchmark

Against the 5 % benchmark adopted in this study (Section 2.8):

- **Baseline:** six of nine buses at or above the benchmark; worst case 8.0781 %.
- **PSO, 1.5 MVA:** one bus above; worst case 5.0285 %.
- **DE, 1.0 MVA:** two buses above; worst case 5.6729 %.
- **DE, 1.5 MVA: all nine compliant;** worst case 4.6866 %.

The successful case clears the benchmark by 0.31 percentage points at the worst bus, a margin of about 6 %. With both harmonic magnitudes at their bound, there is no reserve: any deterioration in operating conditions, any increase in charging demand, or any additional harmonic source would push the system back into non-compliance with no remaining means of response. Compliance is achieved, but without margin.

4.15 Engineering Interpretation

Three conclusions follow for practice.

A storage converter can perform meaningful harmonic compensation alongside its energy function, reducing system-average distortion by roughly a third in this network without additional plant.

The converter must be sized for it. The energy duty alone would have justified around 0.4 MVA; the harmonic duty required 1.5 MVA. Sizing a BESS on energy grounds and then expecting harmonic compensation as a free addition would not have worked here.

And compensator placement matters. The compensator is remote from the distorting load and must act through network impedance, which is the most plausible reason its current requirement so far exceeds published active-filter practice.

4.16 Economic Discussion

The economic consequence follows directly from the sizing result: harmonic capability required a converter roughly 3.75 times the rating that the energy duties alone would justify. Converter cost scales broadly with rating, so this is a real and quantifiable premium.

That premium should be assessed against the alternative, which is a dedicated active power filter with its own converter, enclosure, protection, controls and installation. Upsizing a converter that is being purchased anyway avoids a second complete power-electronic installation, and is plausibly the cheaper route even at 3.75 times the rating – but this study cannot demonstrate that without cost data.

On local availability, the derived 0.4 MVA to 0.5 MVA requirement falls within the ZERO mid tier (100 kW to 800 kW) manufactured by Fakir Technologies Ltd. in Dhaka, which has also deployed a 1 MW industrial unit [31]. The 1.5 MVA rating the harmonic duty requires sits above that tier and would need a custom or imported unit – itself a cost consequence of the finding.

4.17 Social Impact

Poor power quality is not distributed evenly. Distortion propagates to neighbouring buses, so the customers who bear the consequences – equipment overheating, reduced appliance life, flicker – are often not those charging vehicles. A mitigation measure that keeps

distortion within limits at every bus rather than only at the charging station addresses that asymmetry directly, and this is why the objective functions in this work are system-wide rather than local.

For Bangladesh specifically, EV charging infrastructure is being deployed into a network where many consumers already experience variable supply quality. Adding distortion without mitigation would compound an existing problem for households and small businesses that have no means of protecting themselves from it.

4.18 Environmental Impact

The environmental case operates through two mechanisms. Renewable generation supplies 15.97 to 17.38 % of annual energy in the modelled scenarios, displacing conventional generation and its associated emissions. Less directly, reducing harmonic distortion reduces the additional losses and thermal stress that distortion imposes on transformers and cables, which lowers loss energy and extends equipment life.

The magnitude of the second effect is not quantified in this study, and no emissions figure is claimed. Doing so would require loss calculations at both distortion levels and an emissions factor for displaced generation.

4.19 Health, Safety and Ethical Considerations

On safety, harmonic distortion causes additional heating in transformers, cables and motors. Keeping distortion within the benchmark reduces thermal stress and the associated risk of insulation failure. The converter itself must operate within its current rating, which Eq. (3.2) governs; the saturation observed in Section 4.13 means the modelled converter operates at its harmonic current limit, a condition that would require careful thermal verification before implementation.

On ethics, this thesis has aimed to report what the model produced rather than what would have been convenient. Three decisions reflect that. The PSO result that failed the benchmark is reported in full rather than omitted. The seeded initialisation that weakens the algorithm comparison is stated explicitly (Section 4.12). And the disagreement between this study's sizing finding and published practice is presented as a disagreement to be explained (Section 4.13), not suppressed. A result that only confirms expectations is of limited value; one that contradicts them is worth reporting carefully.

4.20 Limitations

1. **Single operating point.** The harmonic study was executed at one operating condition. No reported THD value is an annual statistic.
2. **Uncertainty not propagated into the harmonic study.** The Monte Carlo analysis characterises input variability but is not coupled to the harmonic calculation.
3. **Demand uncertainty not modelled.** EV charging demand was effectively constant across the 300 runs.
4. **Latin hypercube sampling not implemented.** It was considered but not used.
5. **Warm-started DE.** The comparison between algorithms is correspondingly weakened (Section 4.12).
6. **Single random seed.** No repeatability assessment was performed.
7. **Two converter ratings only.** The threshold between them is not located.
8. **Two harmonic orders, one location.** Compensation was confined to the fifth and seventh orders at a single bus.
9. **Frequency-domain model.** Switching behaviour, converter control dynamics and thermal limits are not represented.
10. **Representative charger spectrum.** Absolute THD values are conditional on the spectrum in Table 3.2.
11. **Identical profiles for co-located elements.** Four PV arrays and three chargers share single profiles, removing diversity.
12. **Converter headroom not evaluated numerically.** Optimised magnitudes are percentages of rated current; Eq. (3.2) has not been evaluated in amperes.

4.21 Chapter Summary

The EV charging station produced voltage distortion above the benchmark at six of nine buses, worst at Bus 8 with 8.0781 %. BESS harmonic compensation reduced this substantially, and with a feasibility-first objective differential evolution brought all nine buses below the benchmark, to a maximum of 4.6866 % and an average of 3.5907 %. Both optimised harmonic magnitudes reached their upper bound, and a repeat study at a reduced converter rating confirmed that the binding constraint was converter current rather than search performance.

CHAPTER 5

CONCLUSIONS

5.1 Introduction

This chapter summarises the work completed, states the main findings, assesses them against the research objectives, and sets out what should follow.

5.2 Summary of Completed Work

A generic nine-bus microgrid incorporating PV generation, wind generation, an EV charging station and a battery energy storage system was built in DIgSILENT PowerFactory. An annual 8760-hour dataset was assembled from measured PV and wind generation for Bhola, Bangladesh, and from NREL EVI-Pro Lite charging data. Renewable uncertainty was represented through a 300-run Monte Carlo study using a first-order autoregressive perturbation, preserving the hour-to-hour correlation that independent sampling would discard.

Harmonic load flow established the baseline distortion with the full system connected and BESS compensation disabled. The storage converter was then represented as a balanced, phase-correct harmonic current source, and two optimisation algorithms – particle swarm optimisation and differential evolution – were implemented as PowerFactory Python scripts to tune the magnitude and phase of the injected fifth- and seventh-order components. The study was repeated at a reduced converter rating to identify what limited the result.

5.3 Main Findings

1. **The charging station alone produced non-compliant distortion at most buses.** Baseline voltage THD reached 8.0781 % at Bus 8, with six of nine buses at or above the 5 % benchmark and a system average of 5.3926 %.
2. **Distortion was worst away from the injection point.** The charging station is at Bus 4, but the worst distortion appeared at Bus 8, confirming that harmonic mitigation

in this network must be posed as a system-wide rather than a local problem.

3. **The storage converter achieved compliance at every bus.** Differential evolution with a feasibility-first objective reduced the maximum to 4.6866 % and the average to 3.5907 %, a reduction of 42.0 % in the worst case and 33.4 % on average.
4. **The objective formulation, not the algorithm, determined compliance.** A ratio-based objective left Bus 5 at 5.0285 % despite improving every bus; a feasibility-first objective that treats exceedance as a constraint violation succeeded.
5. **Optimal injection phase was set by network impedance, not by direct opposition.** The selected phases (292.83° and 324.97°) are far from 180° , because the compensator is remote from the distorting load and the compensating current traverses frequency-dependent impedance.
6. **Converter current, not search performance, was the binding constraint.** Both harmonic magnitudes saturated at their 100 % bound. Repeating the study at 1.0 MVA left two buses non-compliant with identical settings, whereas 1.5 MVA succeeded.
7. **Harmonic duty dominated converter sizing.** The energy duties justified approximately 0.4 MVA; harmonic performance required 1.5 MVA. This contradicts published active-filter sizing guidance, and Section 4.13 proposes that compensator location, the system-wide compliance requirement, and a different reference base account for the disagreement.

5.4 Achievement of Research Objectives

Research questions RQ1, RQ2, RQ4 and RQ5 are answered. RQ3, concerning which algorithm performs better, is **not** answered: the two runs used different objectives and different initialisations, and no ranking can be drawn from them.

5.5 Engineering Contribution

The main contribution is the converter-sizing result. Existing work establishes that a storage converter can compensate harmonics; this study quantifies what that capability costs in converter rating for a specific network, and shows that the cost can exceed the rating required for all other duties combined. The finding was obtained by the straightforward expedient of repeating the study at a smaller rating, which is a method other researchers can apply directly.

A second contribution is methodological. Reporting that an optimiser converged at its

Table 5.1: Assessment of each research objective against the results obtained.

No.	Objective	Status	Evidence and qualification
1	Build the PowerFactory model with annual data	Achieved	Sections 3.2 and 3.10
2	Model input uncertainty	Achieved in part	300-run Monte Carlo completed; charging demand not varied; LHS not implemented
3	Quantify baseline distortion	Achieved	Table 4.2; single operating point only
4	Optimise harmonic compensation	Achieved	Section 4.9; all nine buses compliant
5	Compare PSO and DE	Achieved in part	Both implemented; comparison confounded by differing objectives and warm start
6	Relate converter rating to mitigation	Achieved	Table 4.4; two ratings only

bound, rather than presenting the result as an unconstrained optimum, is what made the sizing question visible. A bound-saturated result should be read as an indication that the model, not the search, is the limitation.

5.6 Limitations

The limitations are set out fully in Section 4.20. The three most significant are that the harmonic study was executed at a single operating point and so yields no annual statistic; that the differential evolution run was warm-started, weakening the algorithm comparison; and that the sizing conclusion rests on two ratings rather than a sweep.

5.7 Recommendations

For practitioners designing renewable-supported EV charging microgrids:

1. Size the storage converter for the harmonic duty explicitly if compensation is intended. Sizing on energy grounds and adding harmonic capability afterwards may not be feasible.
2. Do not transfer published active-filter sizing ratios to a converter located away from the distorting load without verification.
3. Assess distortion at every bus, not only at the charging station. The worst bus was not

the injection bus in this network.

4. Formulate the optimisation objective around absolute compliance rather than proportional improvement where a benchmark must be met everywhere.
5. Treat an optimiser that converges at a bound as evidence of a physical limitation and investigate the model rather than the search.

5.8 Future Work

5.8.1 Immediate Extensions

- **Converter rating sweep.** Repeat the DE study at intermediate ratings between 1.0 and 1.5 MVA to locate the compliance threshold and produce a mitigation-versus-rating curve.
- **Fair algorithm comparison.** Run both algorithms under the same feasibility-first objective from unseeded random populations, across multiple random seeds, with mean and variance reported.
- **Numerical headroom evaluation.** Convert the optimised harmonic magnitudes to amperes and evaluate Eq. (3.2) explicitly against the converter rating.

5.8.2 Larger Studies

- **Full annual harmonic simulation.** Execute harmonic load flow at every one of the 8760 hours, applying the hourly PV, wind and charging values, and report mean, maximum, standard deviation, 95th percentile and the count of hours above benchmark. Only this would support genuine annual harmonic statistics, which the present study explicitly does not claim.
- **Uncertainty-coupled harmonic analysis.** Propagate the Monte Carlo scenarios into the harmonic study to obtain a distribution of THD rather than a single value.
- **State-of-charge-aware compensation.** Couple harmonic compensation to battery state of charge and to the fundamental operating point, so that available headroom is computed from actual converter loading at each hour.
- **Explicit converter current constraints.** Enforce Eq. (3.2) as a hard constraint within the optimisation rather than applying a fixed percentage bound.
- **Thermal constraints.** Include converter thermal limits, given that the successful case operates at the harmonic current bound.

- **Adaptive hourly compensation.** Retune injection settings as operating conditions change through the day and year.
- **Multiple compensators and placement optimisation.** Distribute compensation across several units and optimise location, since the present study suggests location strongly affects the current required.
- **Additional harmonic orders.** Extend compensation beyond the fifth and seventh, particularly if the passive filter arrangement changes.
- **Electromagnetic transient modelling.** Represent converter switching and control dynamics to verify that the commanded harmonic currents are realisable by a real controller.
- **Hardware-in-the-loop and measured charger spectra.** Validate against a real converter controller, and replace the representative charger spectrum with measurements from a deployed charger.

5.9 Final Conclusion

An EV charging station in the modelled nine-bus microgrid produced voltage distortion exceeding the 5 % benchmark at six of nine buses, worst at 8.0781 %. Operating the battery storage converter as an active harmonic compensator, with injection magnitude and phase determined by differential evolution under a feasibility-first objective, brought every bus below the benchmark, reducing the worst case to 4.6866 % and the system average by roughly a third.

The more useful result concerns what limited that outcome. Both optimised harmonic magnitudes converged at their upper bound, and repeating the study with a 1.0 MVA converter left two buses non-compliant under otherwise identical conditions. Harmonic mitigation in this system was constrained by available converter current, not by the optimisation. The converter rating required for the harmonic duty exceeded that required for energy storage and peak shaving combined, which means that where a storage converter is intended to provide harmonic compensation as well as energy services, the harmonic function should be treated as a sizing driver in its own right rather than as a capability that comes without cost.

CHAPTER 6

OUTCOME-BASED EDUCATION (OBE) MAPPING

6.1 Introduction

This chapter documents how the project satisfies the Outcome-Based Education requirements of the Department of Electrical and Electronic Engineering, IUT, for EEE 4700/4800: Project and Thesis. It records the Complex Engineering Problem attributes (P1–P7) and Complex Engineering Activity attributes (A1–A5) that the work exhibits, the Knowledge Profile (K3–K8) it draws on, and the Program Outcomes (POs) and Course Outcomes (COs) it addresses. The attribute definitions and tick assignments follow the accepted proposal for this thesis, cross-checked against the outcomes achieved at completion.

6.2 Complex Engineering Problem

The thesis problem satisfies **P1**, together with **P2**, **P4**, and **P7**, and can accordingly be classified as a Complex Engineering Problem.

P1. The problem requires in-depth engineering knowledge spanning power system analysis, renewable energy integration, uncertainty quantification, and engagement with current research literature. A first-principles analytical approach underlies the load-flow and harmonic calculations reported in Chapters 3 and 4.

P2. Conflicting requirements arise between maximising renewable penetration and preserving grid stability, between minimising battery converter cost and providing adequate harmonic-compensation headroom, and between the computational cost of the uncertainty study and the accuracy of its results. Chapter 4 shows how these trade-offs were resolved for the converter sizing adopted in this work.

P4. High-penetration EV charging coordinated with renewable integration under uncertainty is an emerging class of problem that is infrequently encountered in conventional power-system studies, which more often treat generation and load as deterministic.

P7. The problem couples several interdependent sub-problems – EV-charging load modelling, renewable generation variability, battery converter sizing, probabilistic load

Table 6.1: Attributes of the Range of Complex Engineering Problem Solving (P1–P7) addressed by this thesis.

Attr.	Characteristic	Description	Addressed
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3–K8, allowing a fundamentals-based, first-principles analytical approach.	✓
P2	Range of conflicting requirements	Involves wide-ranging or conflicting technical, engineering, and other issues.	✓
P3	Depth of analysis required	Has no obvious solution and requires abstract thinking and originality in analysis to formulate suitable models.	—
P4	Familiarity of issues	Involves infrequently encountered issues.	✓
P5	Extent of applicable codes	Lies outside problems fully encompassed by standards and codes of practice for professional engineering.	—
P6	Extent of stakeholder involvement	Involves diverse groups of stakeholders with widely varying and conflicting requirements.	—
P7	Interdependence	Is a high-level problem comprising many component parts or sub-problems.	✓

flow, and harmonic assessment – each of which affects the others, so that the microgrid had to be analysed as a single integrated system, as carried out in Chapter 3.

6.3 Complex Engineering Activity

A1. The project drew on human resources (supervisor guidance and team collaboration), financial resources (the project budget), computing equipment for simulation, information resources (research literature and the Bhola and NREL datasets described in Chapter 2), and technologies including DIgSILENT PowerFactory and Python-based optimisation tools.

A2. Significant problems arose from the interaction between uncertainty-modelling fidelity and computational feasibility, between power quality compliance and charging service level, and between renewable penetration and grid stability, all of which are

Table 6.2: Attributes of the Range of Complex Engineering Activities (A1–A5) addressed by this thesis.

Attr.	Characteristic	Description	Addressed
A1	Range of resources	Involves the use of diverse resources – people, money, equipment, materials, information, and technologies.	✓
A2	Level of interaction	Requires resolution of significant problems arising from interactions between wide-ranging or conflicting issues.	✓
A3	Innovation	Involves creative use of engineering principles and research-based knowledge in novel ways.	✓
A4	Consequences for society and the environment	Has significant consequences in a range of contexts, characterised by difficulty of prediction and mitigation.	✓
A5	Familiarity	Extends beyond previous experience by applying principles-based approaches.	—

addressed in Chapters 3 and 4.

A3. Innovation is demonstrated through the application of Monte Carlo and Latin Hypercube sampling to EV-charging scenarios, the combination of probabilistic load flow with harmonic analysis, and the use of metaheuristic optimisation (particle swarm and differential evolution) to determine battery-converter harmonic-injection settings, going beyond the deterministic approaches common in the reviewed literature.

A4. The project has consequences for cleaner transportation infrastructure, reduced carbon emissions through renewable integration, and grid reliability for the communities served by the modelled network – outcomes that are difficult to predict with certainty and that motivated the uncertainty-aware treatment adopted in this thesis.

6.4 Knowledge Profile

K4. Modelling EV chargers as nonlinear harmonic-producing loads, analysing harmonic distortion and its propagation through the nine-bus microgrid, and relating that distortion to converter stress required specialist power-system and power-quality knowledge at the forefront of current practice, developed in Chapter 3.

K6. Engineering practice is reflected in the use of the industry-standard DIgSILENT PowerFactory software for power-system analysis, the use of real-world renewable-generation and EV-charging data, and attention to grid-code and power-quality constraints

Table 6.3: Attributes of the Knowledge Profile (K3–K8) addressed by this thesis.

Attr.	Knowledge Profile Attribute	Addressed
K3	A systematic, theory-based formulation of engineering fundamentals required in the discipline.	—
K4	Engineering specialist knowledge providing theoretical frameworks and bodies of knowledge for accepted practice areas, much of it at the forefront of the discipline.	✓
K5	Knowledge that supports engineering design in a practice area.	—
K6	Knowledge of engineering practice (technology) in the practice areas of the discipline.	✓
K7	Comprehension of the role of engineering in society, including ethics, professional responsibility to public safety, and the economic, social, cultural, environmental, and sustainability impacts of engineering activity.	✓
K8	Engagement with selected knowledge in the research literature of the discipline.	✓

throughout Chapters 3 and 4.

K7. The thesis engages with engineering’s role in society by supporting the transition to electric mobility and renewable energy while addressing professional responsibility to public safety through grid stability and power-quality analysis; the environmental and sustainability impact of renewable integration is evaluated directly in Chapter 4.

K8. The methodology draws extensively on the research literature on EV-charging integration, uncertainty quantification, renewable energy systems, and power-quality mitigation, reviewed in Chapter 2, and extends it by incorporating uncertainty-aware renewable-generation behaviour using a full year of real data together with systematic sampling.

6.5 Program Outcomes Addressed

6.6 Course Outcomes and Program Outcome Mapping

Table 6.5 lists the Course Outcomes (COs) for EEE 4700/4800 together with the Program Outcome (PO) each maps to, and indicates which were achieved by the completed thesis and defence.

Table 6.4: Program Outcome (PO) aspects addressed by this thesis.

PO	Statement	Aspect	Addressed
PO3	Design/development of solutions for complex electrical and electronic engineering problems, with appropriate consideration for public health and safety, cultural, societal, and environmental factors.	Societal	✓
		Environmental	✓
PO4	Investigation of complex electrical and electronic engineering problems using research-based knowledge and methods, including design of experiments, analysis and interpretation of data, and synthesis of information.	Design of experiments	✓
		Analysis and interpretation of data	✓
		Synthesis of information	✓
PO6	Reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to professional engineering practice.	Societal	✓
PO7	Understanding and evaluating the sustainability and impact of professional engineering work in societal and environmental contexts.	Environmental	✓
PO8	Application of ethical principles, professional ethics and responsibilities, and the norms of electrical and electronic engineering practice.	Professional ethics and responsibilities	✓
		Norms	✓
PO10	Communication on complex engineering activities: comprehending and writing effective reports and design documentation, making effective presentations, and giving and receiving clear instructions.	Comprehend and write effective reports	✓
		Design documentation	✓
		Make effective presentations	✓
		Give and receive clear instructions	✓
PO11	Engineering management principles and economic decision-making, applied to managing projects individually and within a team.	Engineering management principles	✓
		Manage projects	✓
		Multidisciplinary environments	✓

Table 6.5: Course Outcome (CO) to Program Outcome (PO) mapping, and the outcomes achieved in EEE 4800.

CO	CO Statement	PO	Achieved
CO1	Apply knowledge of electrical and electronic engineering to the solution of complex engineering problems.	PO1	✓
CO2	Identify a contemporary real-life problem related to electrical and electronic engineering by reviewing and analysing existing research works.	PO2	✓
CO3	Determine functional requirements of the problem considering feasibility and efficiency through analysis and synthesis of information.	PO4	✓
CO4	Select a suitable solution and determine its method considering professional ethics, codes, and standards.	PO8	✓
CO5	Adopt modern engineering resources and tools for the solution of the problem.	PO5	✓
CO6	Prepare a management plan and budgetary implications for the solution of the problem.	PO11	✓
CO7	Analyse the impact of the proposed solution on health, safety, culture, and society.	PO6	—
CO8	Analyse the impact of the proposed solution on environment and sustainability.	PO7	—
CO9	Develop a viable solution considering health, safety, cultural, societal, and environmental aspects.	PO3	—
CO10	Work effectively as an individual and as a team member for the accomplishment of the solution.	PO9	✓
CO11	Prepare various technical reports, design documentation, and deliver effective presentations for demonstration of the solution.	PO10	✓
CO12	Recognise the need for continuing education and participation in professional societies and meetings.	PO12	—

CO1. Power-system fundamentals were applied directly by modelling the nine-bus microgrid with PV, wind, and BESS integration, and by running load-flow analysis against the real-world Bhola dataset in Chapter 3; this was extended by harmonic analysis of the EV charging stations, Monte Carlo and Latin Hypercube sampling for uncertainty quantification, and the resulting renewable-penetration and battery-sizing recommendations in Chapter 4.

CO2. The contemporary problem of EV-charging-induced power-quality degradation in renewable-rich microgrids was identified through the literature review in Chapter 2, which highlighted the gap left by studies that treat renewable generation as constant rather than uncertain.

CO3. Functional requirements – the IEEE 519 total harmonic distortion limit, converter and battery sizing, and the load-flow and harmonic-analysis workflow – were determined through the analysis and synthesis of information carried out in Chapter 3.

CO4. DIgSILENT PowerFactory was selected as the simulation platform for its industrial acceptance in power-system engineering practice, and the study was framed throughout around IEEE 519 compliance. Professional ethics were maintained through transparent methodology, honest reporting of assumptions, and avoidance of selective reporting of results, as discussed in Chapter 5.

CO5. Modern engineering resources and tools were adopted throughout, including DIgSILENT PowerFactory for power-system modelling, Python-based Monte Carlo and Latin Hypercube sampling, and the particle-swarm and differential-evolution optimisation routines used for harmonic mitigation in Chapters 3 and 4.

CO6. A structured work plan and budget were prepared, covering system modelling, data collection, and load-flow analysis in the first phase and harmonic analysis, mitigation, thesis writing, and presentation preparation in the second, together with a budget covering software licensing and thesis production costs, as set out in the accepted project proposal.

CO10. Individual contributions – literature review, data collection from the Bhola and NREL sources, and independent simulation tasks – were combined with team collaboration during result analysis, thesis writing, and defence-presentation preparation, reflecting the structured teamwork of the three-member group supervised by Dr. Ashik Ahmed.

CO11. The accepted project proposal, the completed thesis document, and the final defence presentation together satisfy this outcome, covering report writing, design documentation, and effective communication of the uncertainty-based results and harmonic-mitigation recommendations to faculty and peers.

REFERENCES

- [1] J. Chakma, *EVs becoming a common sight on Dhaka roads*, The Daily Star, 25 September 2025, Sep. 2025.
- [2] R. Redwan, M. Hasan, A. Nadia, M. S. Khan, N. A. Chowdhury, and N.-U.-R. Chowdhury, "Design analysis and techno-economic assessment of a photovoltaic-fed electric vehicle charging station at Dhaka–Mawa expressway in Bangladesh," *Energy Conversion and Management: X*, vol. 24, p. 100 737, 2024. doi: 10.1016/j.ecmx.2024.100737.
- [3] M. A. Halim et al., "Integration of renewable energy power plants on a large scale and flexible demand in Bangladesh's electric grid: A case study," *Control Systems and Optimization Letters*, vol. 1, no. 3, pp. 157–168, 2023. doi: 10.59247/csol.v1i3.48.
- [4] D. Alame, M. Azzouz, and N. Kar, "Assessing and mitigating impacts of electric vehicle harmonic currents on distribution systems," *Energies*, vol. 13, no. 12, p. 3257, 2020. doi: 10.3390/en13123257.
- [5] R. T. Kumar and C. C. A. Rajan, "Integration of hybrid PV-wind system for electric vehicle charging: Towards a sustainable future," *e-Prime — Advances in Electrical Engineering, Electronics and Energy*, vol. 6, p. 100 347, 2023. doi: 10.1016/j.prime.2023.100347.
- [6] F. Mišurović and S. Mujović, "Numerical probabilistic load flow analysis in modern power systems with intermittent energy sources," *Energies*, vol. 15, p. 2038, 2022. doi: 10.3390/en15062038.
- [7] G. E. Constante-Flores and M. S. Illindala, "Data-driven probabilistic power flow analysis for a distribution system with renewable energy sources using Monte Carlo simulation," *IEEE Transactions on Industry Applications*, vol. 55, no. 1, pp. 174–184, 2019. doi: 10.1109/TIA.2018.2865167.
- [8] M. Ovaskainen, J. Öörni, and A. Leinonen, "Superposed control strategies of a BESS for power exchange and microgrid power quality improvement," in *IEEE International Conference on Environment and Electrical Engineering (EEEIC)*, 2019. doi: 10.1109/EEEIC.2019.8783764.
- [9] M. P. Behera and P. K. Ray, "Three-phase grid connected bi-directional charging system to control active and reactive power with harmonic compensation," *International Journal of Emerging Electric Power Systems*, 2019. doi: 10.1515/IJEEPS-2018-0259.

- [10] S. Pranith, S. Kumar, B. Singh, and T. S. Bhatti, "Design and control of a multifunctional grid-connected battery energy storage with enhanced performance using SOGI," *Journal of The Institution of Engineers (India): Series B*, 2019. DOI: 10.1007/S40031-019-00382-3.
- [11] United States Patent and Trademark Office, *Hybrid parallel active/passive filter system with dynamically variable inductance*, U.S. Patent 5,757,099, 1998.
- [12] X.-J. Dong, J.-N. Shen, Z.-F. Ma, and Y.-J. He, "Stochastic optimization of integrated electric vehicle charging stations under photovoltaic uncertainty and battery power constraints," *Energy*, vol. 314, p. 134 163, 2025. DOI: 10.1016/j.energy.2024.134163.
- [13] D. C. Tairo, J. A. A. Silva, J. C. López, and M. J. Rider, "Implementation of a microgrid energy management system considering fair EV charging, uncertainties and contingencies: A multi-objective approach," *Applied Energy*, vol. 396, p. 126 242, 2025. DOI: 10.1016/j.apenergy.2025.126242.
- [14] M. T. Kenari and A. Ozdemir, "Simultaneous impacts of correlated photovoltaic systems and fast electric vehicle charging stations on the operation of active distribution grids," *Sustainable Energy, Grids and Networks*, vol. 40, p. 101 583, 2024. DOI: 10.1016/j.segan.2024.101583.
- [15] A. Srivastava and S. Saravanan, "Harmonic mitigation using optimal active power filter for the improvement of power quality for an electric vehicle charging station," *e-Prime — Advances in Electrical Engineering, Electronics and Energy*, vol. 8, p. 100 527, 2024. DOI: 10.1016/j.prime.2024.100527.
- [16] T. I. Rahman, "Impact of solar generation on IEEE 9-bus system," *Electronic Theses and Dissertations*, 3818, Ph.D. dissertation, University of Maine, 2023.
- [17] A. Khan, B. Hussain, A. Wahyudie, and R. A. Khalil, "Stability improvement of renewable energy integrated power network through coordination of PSS, ISFCL, and STATCOM," *IEEE Access*, vol. 13, pp. 133 510–133 524, 2025. DOI: 10.1109/ACCESS.2025.3592940.
- [18] K. Srilakshmi et al., "Optimal design of solar/wind/battery and EV fed UPQC for power quality and power flow management using enhanced most valuable player algorithm," *Frontiers in Energy Research*, vol. 11, p. 1 342 085, 2024. DOI: 10.3389/fenrg.2023.1342085.
- [19] S. M. Abd Elazim et al., "Enhancing stability and power quality in electric vehicle charging stations powered by hybrid energy sources through harmonic mitigation and load management," *Scientific Reports*, vol. 15, p. 28 077, 2025. DOI: 10.1038/s41598-025-14143-4.

- [20] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proceedings of the IEEE International Conference on Neural Networks*, 1995, pp. 1942–1948.
- [21] R. Storn and K. Price, "Differential evolution — a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, no. 4, pp. 341–359, 1997.
- [22] IEEE, *IEEE recommended practice and requirements for harmonic control in electric power systems*, IEEE Std 519.
- [23] N. Jewell et al., *Modeling and simulation of a utility-scale battery energy storage system*, <https://www.researchgate.net/publication/338941863>, 2019.
- [24] Ekistica, *Ramp rate control for PV plant integration: Experience from Karratha airport's hybrid power station*, 36th European Photovoltaic Solar Energy Conference and Exhibition, for the Australian Renewable Energy Agency.
- [25] "Sizing of energy storage systems for ramp rate control of photovoltaic strings," *Renewable Energy*, 2022.
- [26] G. Koolman et al., "Optimal battery energy storage system sizing for demand charge management in EV fast charging stations," in *IEEE*, 2021.
- [27] P. Denholm, W. Cole, and N. Blair, "Moving beyond 4-hour Li-ion batteries: Challenges and opportunities for long(er)-duration energy storage," National Renewable Energy Laboratory, Tech. Rep. NREL/TP-6A40-85878, 2023.
- [28] National Renewable Energy Laboratory, *Utility-scale battery storage*, 2023 Annual Technology Baseline, 2023.
- [29] *KU-HMG2: A dataset of a hybrid micro-grid coupled with photovoltaic panel, wind turbine and natural gas generator (Bhola area, Bangladesh)*, Mendeley Data.
- [30] National Renewable Energy Laboratory, *EVI-Pro lite: Electric vehicle charging infrastructure projection tool*.
- [31] Fakir Technologies Ltd., *ZERO battery energy storage system product line*, <https://www.fakirtechnologies.com/bess-supplier-bangladesh/>, Dhaka, Bangladesh.

APPENDIX A

POWERFACTORY PYTHON SCRIPTS

Both optimisation routines were implemented as PowerFactory Python scripts (ComPython) executing inside the PowerFactory environment, so that the harmonic load flow command and the bus terminal objects are addressed directly.

A.1 Differential Evolution Script

This is the script that produced the final reported result (Section 4.9). The complete listing should be inserted from the project file `DE_code.txt`.

```
1 import random
2 import powerfactory
3
4 app = powerfactory.GetApplication()
5 script = app.GetCurrentScript()
6 hsrc = script.hsrc
7 hldf = app.GetFromStudyCase("ComHldf")
8
9 # --- collect the nine bus terminal objects -----
10 buses = []
11 for i in range(1, 10):
12     obj = app.GetCalcRelevantObjects("Bus %d.ElmTerm" % i)
13     if not obj:
14         raise Exception("Bus %d not found." % i)
15     buses.append(obj[0])
16
17 TARGET = 4.99      # per-bus THD target (%)
18 DIM = 4            # [mag5, phase5, mag7, phase7]; H11 fixed at 0
19 POP_SIZE = 60
20 MAX_GEN = 100
21 CR = 0.90
22 F_MIN, F_MAX = 0.45, 0.90
23 random.seed(42)
24
25 def set_hsrc(x):
26     hsrc.kbal = [x[0], x[2], 0.0]
27     hsrc.cbal_phase = [x[1], x[3], 0.0]
28
29 def run_hlf():
30     hldf.iopt_allfrq = 1
31     if hldf.Execute() != 0:
32         raise Exception("Harmonic load flow failed.")
33     return [float(b.GetAttribute("m:THD")) for b in buses]
34
35 # --- baseline: compensation disabled -----
```

```

36 hsrc.kbal      = [0.0, 0.0, 0.0]
37 hsrc.cbal_phase = [0.0, 0.0, 0.0]
38 baseline = run_hlf()
39
40 # --- lexicographic feasibility-first objective -----
41 def objective(thd):
42     excess = [max(0.0, v - TARGET) for v in thd]
43     return (max(excess),
44             sum(1 for e in excess if e > 0.0),
45             sum(e * e for e in excess),
46             max(thd),
47             sum(thd) / len(thd))
48
49 # --- shortest-arc angular difference -----
50 def circ_diff(a, b):
51     return (a - b + 180.0) % 360.0 - 180.0
52
53 # [ mutation / binomial crossover / greedy selection / coordinate
54 #   polish -- see the full listing in DE_code.txt ]

```

Listing A.1: Differential evolution harmonic optimisation (abridged — insert full listing from DE_code.txt)

A.2 Particle Swarm Optimisation Script

```

1 DIM      = 6      # [mag5, ph5, mag7, ph7, mag11, ph11]
2 N_PARTICLES = 24
3 N_ITERATIONS = 45
4 W_START, W_END = 0.9, 0.35
5 C1, C2      = 1.6, 1.8
6 VMAX_FRAC   = 0.25
7 random.seed(42)
8
9 def objective(thd):
10     """Worst-case THD ratio against baseline, tie-broken by the mean."""
11     ratios = [thd[i] / baseline[i] if baseline[i] > 1e-9 else 0.0
12               for i in range(N_BUSES)]
13     return (max(ratios), sum(ratios) / len(ratios))
14
15 # [ velocity update with circular handling of phase dimensions,
16 #   position clamping, pattern-search polish -- see PSO_code.txt ]

```

Listing A.2: Particle swarm harmonic optimisation (abridged — insert full listing from PSO_code.txt)

APPENDIX B

DATASET INFORMATION

B.1 Source Datasets

PV and wind generation. KU-HMG2, a dataset of a hybrid microgrid coupled with photovoltaic panels, a wind turbine and a natural gas generator for the Bhola area, Bangladesh [29]. Hourly resolution over a full year.

EV charging demand. NREL EVI-Pro Lite [30], which generates charging demand profiles from vehicle usage patterns.

B.2 Profiles Applied in the PowerFactory Model

Table B.1: Annual profiles applied to model elements.

File	Element	Peak (kW)	Annual (kWh)
PVA1–PVA4	PV arrays, Bus 6	307.03	134,598
WTG1	Wind generator, Bus 5	172.08	43,144
Charger 1–3	EV chargers, Bus 4	52.31	127,750

Note that the four PV files are identical to one another, as are the three charger files. Each group represents co-located elements sharing a single profile, which removes diversity between them (Sections 3.4 and 3.6).

B.3 Data Preparation Notes

All series are hourly with 8760 points. Timestamps in the source files carry differing calendar years; these were aligned to a common index before use, since only the hourly sequence is relevant.

APPENDIX C

HARMONIC PARAMETERS

C.1 EV Charger Harmonic Source

Table C.1: Charger harmonic current spectrum as configured in PowerFactory.

Order	Magnitude (% of reference current)
3	2.8352
5	2.6949
7	1.5853
9	1.3670

Charger rated current is approximately 0.2510 kA. The source was originally inspected as an IEC 61000 type; because IEC spectra specify magnitude without phase, and phase is essential to active cancellation, the compensation study uses a balanced, phase-correct representation with harmonic currents referred to rated current.

C.2 Optimised BESS Compensator Settings

Table C.2: Optimised harmonic injection settings for each case.

Case	5th harmonic		7th harmonic	
	Mag. (%)	Phase (°)	Mag. (%)	Phase (°)
PSO, 1.5 MVA	90.279	334.90	98.714	320.50
DE, 1.5 MVA	100.000	292.83	100.000	324.97
DE, 1.0 MVA	100.000	347.56	100.000	333.07

The eleventh harmonic is zero in all reported cases.

C.3 Superseded Development-Stage Settings

Retained for completeness; these are not thesis results. An early manual estimate gave $H5 = 53.898\%$ and $H7 = 31.706\%$, and a coordinate search gave $H5 = 40.4235\%$ at 344° and $H7 = 49.6325\%$ at 330° . These belong to an earlier model build with a different baseline and must not be compared against the results in Chapter 4.

APPENDIX D

MONTE CARLO SCENARIO GENERATION

D.1 Procedure

1. Load the deterministic annual PV, wind, base-load and EV series.
2. For each of 300 runs, draw a first-order autoregressive noise sequence and apply it multiplicatively to the PV and wind series.
3. Recompute total load and net load for the perturbed series.
4. Record per-run peak, mean and 95th-percentile statistics.
5. Export five representative scenarios in full alongside the deterministic base case.

D.2 Results Across 300 Runs

Table D.1: Statistics across the 300 Monte Carlo runs.

Quantity	Minimum	Mean	Maximum
PV peak (kW)	174.48	253.74	533.38
Wind peak (kW)	99.99	250.80	858.73
EV peak (kW)	309.81	309.82	309.83
Net load peak (kW)	307.12	315.70	325.15
Net load mean (kW)	114.25	115.31	116.62
Night-time PV (kW)	0.00	0.00	0.00

The near-zero spread in EV peak confirms that charging demand was not varied across runs; this study addresses renewable uncertainty only. The zero night-time PV row is a validation check confirming the multiplicative perturbation introduced no generation where the underlying series was zero.

APPENDIX E

SUPPLEMENTARY RESULTS

E.1 PSO Convergence Trace

The complete trace of the best worst-case THD ratio over 45 iterations is held in `data/pso_convergence.csv` and plotted in Fig. 4.1. The ratio falls from 0.837,08 at initialisation to 0.804,35 at the final iteration.

E.2 DE Generation Log

Table E.1: Differential evolution generation log for the 1.5 MVA case.

Generation	Max THD (%)	Avg THD (%)
1	4.8865	3.4036
2	4.8865	3.4036
3	4.8865	3.4036
4	4.8865	3.4036
5	4.8377	3.5226
6	4.8012	3.6124
7	4.8012	3.6124
8	4.6954	3.5804

The run terminated at generation 8 when the success counter reached its threshold of eight consecutive feasible generations. The maximum THD of the best individual is already below the 4.99 % target at generation 1, which reflects the seeded initial population described in Section 3.17.1 and is the basis for the qualification in Section 4.12. The final reported value of 4.6866 % results from the coordinate polish applied after the main loop.

E.3 Full Monte Carlo Summary

Per-run statistics for all 300 runs are held in the project data file `MC_Summary_VAR_AR1_300runs.csv`; summary statistics appear in Table D.1.